

UNPOISON

EARTH • AIR • WATER

SA's total HHPs = 224
SA's banned HHP's = 29
Legally in use including restricted = 195
EU's not approved HHPs = 135
EU not approved and banned in SA = 19
EU not approved and legal in SA = 116
Coformulant HHPs = 16 (17 - 1 banned)

KEY	
X	SA's Banned HHPs
XX	EU's 'not approved' (formerly referred to as banned)
Y	Co-formulants

		Criteria 1	Criteria 2	Criteria 3	Criteria 4	Criteria 5	Criteria 6	Criteria 7	Criteria 8	COMMENTS	Hazard classes and Categories *	Hazard pictograms	GHS classification	ECHA classification
Total = 224 of 2688 registered pesticides														
CAS NO.		WHO 1a & 1b	Carcinogenicity 1A & 1B (GHS)	Mutagenicity 1A & 1B (GHS)	Reproductive toxicity 1A & 1B (GHS)	Stockholm	Rotterdam	Montreal	High incidences of severe or irreversible adverse effects on human health and environment					
106-93-4	XXY		1					1	1	CO-FORMULANT Chemical subject to PIC regulations -ANNEX III Highly volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable Mobile in soils Toxic to mammals Carcinogen Genotoxic Neurotoxicant Respiratory tract, skin & eye irritant Harmful if swallowed Probable cardiovascular, nervous system, liver and kidney toxicant	Flam. Liq. 2 (100%) Acute Tox. 4 (100%) Skin Irrit. 2 (100%) Eye Irrit. 2A (100%) Acute Tox. 3 (46.26%) STOT SE 3 (100%) Carc. 1B (100%)		H225 (100%): Highly Flammable liquid and vapor [Danger Flammable liquids] H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H315 (100%): Causes skin irritation [Warning Skin corrosion/irritation] H319 (100%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H331 (46.26%): Toxic if inhaled [Danger Acute toxicity, inhalation] H335 (100%): May cause respiratory irritation [Warning Specific target organ toxicity, single exposure; Respiratory tract irritation] H350 (100%): May cause cancer [Danger Carcinogenicity]	Flam. Liq. 2 H225 Acute Tox. 4 H302 Skin Irrit. 2 H315 Eye Irrit. 2 H319 STOT SE 3 H335 Carc. 1B H350
	Y		1		1				1	CO-FORMULANT Drinking water contaminant Non-persistent in soils Moderately mobile in soils Low leachability Toxic to mammals & aquatic organisms Carcinogen Reproductive / developmental toxicant Neurotoxicant Respiratory tract, skin & eye irritant Possible liver, kidney, cardiovascular and blood toxicant	Acute Tox. 3 (61.62%) Acute Tox. 3 (61.62%) Eye Irrit. 2 (63.31%) Acute Tox. 4 (70.59%) Muta. 2 (61.62%) Carc. 1B (100%) Repr. 1B (67.79%) STOT RE 1 (72.27%) STOT RE 2 (56.02%) Aquatic Chronic 2 (61.62%)		H301+H311 (47.62%): Toxic if swallowed or in contact with skin [Danger Acute toxicity, oral; acute toxicity, dermal] H301 (61.62%): Toxic if swallowed [Danger Acute toxicity, oral] H311 (61.62%): Toxic in contact with skin [Danger Acute toxicity, dermal] H319 (63.31%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H332 (70.59%): Harmful if inhaled [Warning Acute toxicity, inhalation] H341 (61.62%): Suspected of causing genetic defects [Warning Germ cell mutagenicity] H350 (100%): May cause cancer [Danger Carcinogenicity] H360 (67.79%): May damage fertility or the unborn child [Danger Reproductive toxicity] H372 (72.27%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H373 (56.02%): May causes damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H411 (61.62%): Toxic to aquatic life with long lasting effects [Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 H301 Acute Tox. 3 H311 Acute Tox. 3 H332 Carc. 1B H350 Eye Irrit. 2A H319 STOT SE 3 H373 STOT RE 2 H372 Repr. 1B H360 Aquatic Chronic 1 H411

UnPoison South Africa's Highly Hazardous Pesticide Database vs EU's According to JMPM Criteria (FAO/WHO), GHS and ECHA Classification Systems - February 2025

78-87-5	XXY	1,2-Dichloropropane (Propylene dichloride)							CO-FORMULANT Severe water pollutant Highly soluble in water Highly volatile If applied directly to plants or soil, drift is a concern & mitigation is advisable / Highly volatile. Not persistent in soil or aquatic systems Mobile in soil Toxic to mammals, birds, honeybees & earthworms Toxic to aquatic organisms Public Exposure Routes - Low risk via air, water or food due to usage pattern (WHO) Occupational Exposure Routes - Potentially via inhalation and skin absorption Personal protection clothing and equipment must be worn when re-entering treated areas Genotoxic Reproductive / developmental toxicant Respiratory tract, skin & eye irritant Skin sensitiser Aspiration hazard Highly toxic Possible urinary, liver and kidney toxicant	Flam. Liq. 2 (100%) Acute Tox. 4 (90.8%) Acute Tox. 4 (74.4%) Carc. 1B (62.4%)		H225 (100%): Highly Flammable liquid and vapor [Danger Flammable liquids] H302 (90.8%): Harmful if swallowed [Warning Acute toxicity, oral] H332 (74.4%): Harmful if inhaled [Warning Acute toxicity, inhalation] H350 (62.4%): May cause cancer [Danger Carcinogenicity]	Flam. Liq. 2 H225 Acute Tox. 4 H302 Acute Tox. 4 H332 Carc. 1B H350
542-75-6	XXY	1,3-dichloropropene						1	CO-FORMULANT Severe water pollutant Chemical subject to PIC regulations Halogenated hydrocarbon used as a soil fumigant Highly volatile If applied directly to plants or soil, drift is a concern & mitigation is advisable Persistent & mobile in soils Moderately toxic to mammals Moderately toxic to birds, honeybees, earthworms and most aquatic organisms Occupational exposure routes - Potentially via inhalation and skin absorption Aspiration hazard	Flam. Liq. 3 (98.44%) Acute Tox. 3 (100%) Asp. Tox. 1 (89.06%) Acute Tox. 3 (89.45%) Acute Tox. 4 (10.55%) Skin Irrit. 2 (100%) Skin Sens. 1 (100%) Eye Irrit. 2A (100%) Acute Tox. 3 (35.94%) Acute Tox. 4 (62.5%) STOT SE 3 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (91.41%)		H226 (98.44%): Flammable liquid and vapor [Warning Flammable liquids] H301+H311+H331 (14.06%): Toxic if swallowed, in contact with skin or if inhaled [Danger Acute toxicity, oral; acute toxicity, dermal; acute toxicity, inhalation] H301+H311 (10.94%): Toxic if swallowed or in contact with skin [Danger Acute toxicity, oral; acute toxicity, dermal] H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H304 (89.06%): May be fatal if swallowed and enters airways [Danger Aspiration hazard] H311 (89.45%): Toxic in contact with skin [Danger Acute toxicity, dermal] H312 (10.55%): Harmful in contact with skin [Warning Acute toxicity, dermal] H315 (100%): Causes skin irritation [Warning Skin corrosion/irritation] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H319 (100%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H331 (35.94%): Toxic if inhaled [Danger Acute toxicity, inhalation] H332 (62.5%): Harmful if inhaled [Warning Acute toxicity, inhalation] H335 (100%): May cause respiratory irritation [Warning Specific target organ toxicity, single exposure; Respiratory tract irritation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (91.41%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Flam. Liq. 3 H226 Acute Tox. 3 H301 Acute Tox. 3 H311 Skin Irrit. 2 H315 Eye Irrit. 2 H319 Skin Sens. 1 H317 Acute Tox. 4 H312 Asp. Tox. 1 H304 STOT SE 3 H335 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
		1-methylcyclopropene						1	Potential groundwater pollutant Highly toxic to earthworms Possible reproductive/developmental toxicant Plant growth regulator Highly volatile If applied directly to plants or soil, drift is a concern & mitigation is advisable Persist in aquatic systems Potential bioaccumulation Moderately toxic to fish and algae Highly toxic to birds & earthworms Toxic to bees Low ADI / ARID / AOEL Small risk from digestion of treated food commodities Possibly slight effects on some haematological parameters	Not classified under GHS		not classified under GHS	Acute Tox.1 H330

UnPoison South Africa's Highly Hazardous Pesticide Database vs EU's According to JMPM Criteria (FAO/WHO), GHS and ECHA Classification Systems - February 2025

[illegible]

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[illegible]

UnPoison South Africa's Highly Hazardous Pesticide Database vs EU's According to JMPM Criteria (FAO/WHO), GHS and ECHA Classification Systems - February 2025

[illegible]

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135158-54-2	XX	Acibenzolar-S-methyl								1	1	Volatile Persistent in water Slightly mobile If applied directly to plants, drift is a concern & mitigation is advisable Slightly mobile in soils but non-persistent Potential for bioaccumulation Toxic to mammals Toxic to birds, honeybees, earthworms and most aquatic organisms Public exposure routes - Minimal risk of dietary exposure Genotoxic Reproductive / developmental toxicant Respiratory tract irritant Possible skin sensitiser Possible skin & eye irritant Possible skin sensitiser Weak evidence of clastogenicity	Skin Irrit. 2 (100%) Skin Sens. 1 (100%) Eye Irrit. 2 (100%) STOT SE 3 (100%) Repr. 1B (40.58%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H315 (100%): Causes skin irritation [Warning Skin corrosion/irritation] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H319 (100%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H335 (100%): May cause respiratory irritation [Warning Specific target organ toxicity, single exposure; Respiratory tract irritation] H360D (40.58%): May damage the unborn child [Danger Reproductive toxicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Skin Irrit. 2 H315 Eye Irrit. 2 H319 Skin Sens. 1 H317 STOT SE 3 H335 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
101007-06-1	XX	Acrinathrin								1	1	Semi-volatile persistent in soils Will rapid degrade in aqueous systems Potential for bioaccumulation and most aquatic species toxic to birds and earthworms exposure routes - Risks identified - protective clothing and equipment advised Possible kidney and blood toxicant Moderately Non-mobile in soils Potential Highly toxic to bees Occupational Toxic Eye irritant	Acute Tox. 4 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H332 (100%): Harmful if inhaled [Warning Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4H332 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
15972-60-8	XX	Alachlor								1	1	Potential groundwater contaminant subject to PIC regulations If applied directly to plants or soil, drift is a concern & mitigation is advisable Mobile in soils Low potential for bioaccumulation Toxic to mammals, birds, most aquatic organisms, honeybees and earthworms Occupational exposure may occur through inhalation and dermal contact Eye irritant Possible carcinogen Possible reproductive / developmental toxicant Competitive binding to estrogen and progesterone receptors Possible skin sensitiser & irritant kidney toxicant Causes irreversible eye damage Chemical Volatile Non-persistent in soil & water Possible EDC - Probable lung and	Acute Tox. 4 (100%) Skin Sens. 1 (100%) Carc. 2 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 H302 Skin Sens. 1 H317 Carc. 2 H351 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
116-06-3	XXX	Aldicarb								1	1	Potential groundwater contaminant Marine Pollutant Non-volatile Persistent in soil & Non-mobile Low leachability Highly toxic to birds and honey bees Toxic to aquatic environment EDC Neurotoxin	Acute Tox. 1 (100%) Acute Tox. 1 (34.52%) Acute Tox. 3 (65.48%) Acute Tox. 1 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300+H310+H330 (33.33%): Fatal if swallowed, in contact with skin or if inhaled [Danger Acute toxicity, oral; acute toxicity, dermal; acute toxicity, inhalation] H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H310 (34.52%): Fatal in contact with skin [Danger Acute toxicity, dermal] H311 (65.48%): Toxic in contact with skin [Danger Acute toxicity, dermal] H330 (100%): Fatal if inhaled [Danger Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 H300 Acute Tox. 3 H310 Acute Tox. 2 H311 Aquatic Acute 1 H400 Aquatic Chronic 1 H410

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	Aldrin								POP - regulated by Stockholm Convention Rotterdam Convention (Class O) - Subject to PIC regulations- ANNEX III Marine Pollutant Low solubility in water Moderately volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable Non-mobile in water Low leachability Bioaccumulative Highly toxic for mammals, birds & honey bees Toxic to earthworms Highly toxic to fish & aquatic invertebrates Occupational Exposure Routes - May be absorbed through intact skin Neurotoxicant Highly toxic by ingestion US EPA - probable human carcinogen Endocrine issues - Competitive binding to androgen receptors Possible Carcinogen, Endocrine disruptor, Reproductive / developmental toxicant & Respiratory tract, skin & eye irritant	Acute Tox. 2 (60%) Acute Tox. 3 (40%) Acute Tox. 1 (70%) Acute Tox. 3 (30%) Carc. 2 (100%) STOT RE 1 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300 (60%): Fatal if swallowed [Danger Acute toxicity, oral] H301+H311 (10%): Toxic if swallowed or in contact with skin [Danger Acute toxicity, oral; acute toxicity, dermal] H301 (40%): Toxic if swallowed [Danger Acute toxicity, oral] H310 (70%): Fatal in contact with skin [Danger Acute toxicity, dermal] H311 (30%): Toxic in contact with skin [Danger Acute toxicity, dermal] H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 H301 Acute Tox. 3 H311 Carc. 2 H351 STOT RE 1 H372 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
X	Alpha-cypermethrin					1	1		Serious water pollutant pollutant Highly toxic to bees Moderately persistent in soil and water Highly toxic to mammals Highly toxic to most aquatic organisms Reproductive / developmental toxicant tract irritant Possible neurotoxicant toxic by ingestion	Acute Tox. 3 (25.9%) Acute Tox. 4 (74.1%) Acute Tox. 4 (67.86%) STOT SE 3 (100%) STOT RE 2 (31.95%) Aquatic Acute 1 (88.66%) Aquatic Chronic 1 (100%)		H301 (25.9%): Toxic if swallowed [Danger Acute toxicity, oral] H302+H332 (10.21%): Harmful if swallowed or if inhaled [Warning Acute toxicity, oral; acute toxicity, inhalation] H302 (74.1%): Harmful if swallowed [Warning Acute toxicity, oral] H332 (67.86%): Harmful if inhaled [Warning Acute toxicity, inhalation] H335 (100%): May cause respiratory irritation [Warning Specific target organ toxicity, single exposure; Respiratory tract irritation] H373 (31.95%): May causes damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H400 (88.66%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 H301 STOT SE 3 H335 STOT RE 2 H373 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
67375-30-8	XX	Aluminium phosphide							Highly toxic to bees spontaneously and release toxic flammable gases Potential for bio-concentration volatile & non-persistent aquatic environment mammals and honeybees Toxic to aquatic invertebrates, algae and earthworms Possible reproductive / developmental toxicant respiratory tract & eye irritant May cause dermatitis Heart and lung toxicant	Water-react. 1 (100%) Acute Tox. 2 (100%) Acute Tox. 3 (20%) Acute Tox. 1 (11.11%) Aquatic Acute 1 (100%)		H260 (100%): In contact with water releases flammable gases which may ignite spontaneously [Danger Substances and mixtures which in contact with water, emit flammable gases] H300+H330 (11.11%): Fatal if swallowed or if inhaled [Danger Acute toxicity, oral; acute toxicity, inhalation] H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H311 (20%): Toxic in contact with skin [Danger Acute toxicity, dermal] H330 (11.11%): Fatal if inhaled [Danger Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard]	Water-react. 1 H260 Acute Tox. 2 H300 Acute Tox. 3 H311 Acute Tox. 1 H330 Aquatic Acute 1 H400
	Ametryn								Chemical subject to PIC regulations Persistent in soils and water Toxic to aquatic organisms	Acute Tox. 4 (100%) Acute Tox. 3 (18.97%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H331 (18.97%): Toxic if inhaled [Danger Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 H302 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
834-12-8	XX	Amisulbrom					1	1	Volatile mobility Persistent in soil systems Potential for bioconcentration birds and honeybees earthworms & aquatic organisms Eye irritant Reproductive / developmental toxicant Possible carcinogen toxicant	Eye Irrit. 2 (43.51%) Acute Tox. 4 (27.48%) Carc. 2 (43.51%) Aquatic Acute 1 (99.24%) Aquatic Chronic 1 (70.99%)		H319 (43.51%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H332 (27.48%): Harmful if inhaled [Warning Acute toxicity, inhalation] H351 (43.51%): Suspected of causing cancer [Warning Carcinogenicity] H400 (99.24%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (70.99%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Eye Irrit. 2 H319 Carc. 2 H351 Aquatic Acute 1 H400 Aquatic Chronic 1 H410

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330089-61-1	XX	Amitraz								1	Volatile directly to plants, drift is a concern & mitigation Slightly mobile bioaccumulation Non-persistent in soils Toxic to mammals, birds, earthworms , honeybees & beneficial insects (<i>Typhlodromus pyri</i>) to aquatic organisms exposure route - Occupational exposure may occur through inhalation and dermal contact Skin irritant Possible carcinogen EDC reproductive / developmental toxicant Ingestion may cause slow heart beat, low blood pressure, sedation and low body temperature	If applied Potential for CAN'T BE USED ON HORSES Low Leachability Neurotoxicant Skin sensitiser Possible	Acute Tox. 4 Skin Sens. 1 STOT RE 2 Aquatic Acute 1 Aquatic Chronic 1 Carc. 1b.		H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H311 (27.54%): Toxic in contact with skin [Danger Acute toxicity, dermal] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H373 (100%): May causes damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 Skin Sens. 1 STOT RE 2 Aquatic Acute 1 Aquatic Chronic 1 H410
		Arsen and its compounds								1	Potential groundwater pollutant Very toxic to aquatic organisms	Toxic to bees Inhalation risk	Acute Tox. 3 (100%) Acute Tox. 3 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H331 (100%): Toxic if inhaled [Danger Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 Acute Tox. 3 Aquatic Acute 1 Aquatic Chronic 1 H410
		Arsenic acid								1	Potential groundwater pollutant Chemical Very toxic to aquatic organisms reproductive and developmental toxicant	Toxic to bees PAN Bad Actor Potential	Acute Tox. 3 (100%) Acute Tox. 4 (78.57%) Skin Corr. 1C (78.57%) Eye Dam. 1 (78.57%) Acute Tox. 3 (100%) Carc. 1A (100%) Repr. 2 (78.57%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H301+H331 (78.57%): Toxic if swallowed or if inhaled [Danger Acute toxicity, oral; acute toxicity, inhalation] H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H312 (78.57%): Harmful in contact with skin [Warning Acute toxicity, dermal] H314 (78.57%): Causes severe skin burns and eye damage [Danger Skin corrosion/irritation] H318 (78.57%): Causes serious eye damage [Danger Serious eye damage/eye irritation] H331 (100%): Toxic if inhaled [Danger Acute toxicity, inhalation] H350 (100%): May cause cancer [Danger Carcinogenicity] H361 (78.57%): Suspected of damaging fertility or the unborn child [Warning Reproductive toxicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 Skin Corr. 1C Eye Dam. 1 Acute Tox. 3 Repr. 2 Aquatic Acute 1 Aquatic Chronic 1 H301
3337-71-1	XX	Asulam								1	Chemical subject to PIC regulations Very volatile & mobile in soil Persistent in water Risk of bioaccumulation to birds, honeybees and earthworms Skin, eye and respiratory tract irritant developmental toxicant Skin & eye irritant Respiratory tract irritant Kidney, thyroid and blood toxicant.	Very volatile & mobile Persistent in water Toxic to Toxic to aquatic organisms Reproductive /	Acute Tox. 4		H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral]	Acute Tox. 4 H302
1912-24-9	XX	Atrazine								1	Groundwater contaminant regulations Moderately toxic to bees and earthworms Very toxic to aquatic organisms Chemical EDC neurotoxicant	Subject to PIC Water pollutant PAN Bad Actor Possible carcinogen Possible	Skin Sens. 1 (100%) Eye Irrit. 2 (20%) STOT RE 2 (100%) Aquatic Acute 1 (98.95%) Aquatic Chronic 1 (100%)		H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H319 (20%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H373 (100%): May causes damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H400 (98.95%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Skin Sens. 1 STOT RE 2 Aquatic Acute 1 Aquatic Chronic 1 H410

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86-50-0	XX	Azinphos-methyl	1							Chemical subject to PIC regulations - PIC candidate Potential groundwater contaminant Marine Pollutant Chemical toxic to bees and birds Acetyl cholinesterase inhibitor PAN Bad Actor Highly toxic to aquatic Neurotoxicant	Acute Tox. 2 (100%) Acute Tox. 3 (90.16%) Skin Sens. 1 (100%) Acute Tox. 2 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H311 (90.16%): Toxic in contact with skin [Danger Acute toxicity, dermal] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H330 (100%): Fatal if inhaled [Danger Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 Acute Tox. 3 Skin Sens. 1 Acute Tox. 2 Aquatic Acute 1 Aquatic Chronic 1 H311 H317 H330 H400 Aquatic Chronic 1 H410
17804-35-2	XX	Benomyl					1			Chemical subject to PIC regulations (some formulations) Marine Pollutant Non-volatile mobile Persistent in water systems persistent in soil systems earthworms and most aquatic organisms Occupational exposure route - Occupational exposure may occur through inhalation and dermal contact Reproductive / developmental toxicant Genotoxic Unspecified genotoxicity type (miscellaneous data source) Respiratory tract & skin irritant skin sensitiser May cause eye defects in new borns - anophthalmia May cause contact dermatitis and may be a skin sensitiser US EPA - possible human carcinogen Slightly Low volatility Moderately Toxic to birds, honeybees, EDC - Endocrine issues - Increase of estrogen production and aromatase activity Possible carcinogen	Acute Tox. 4 (29.84%) Skin Irrit. 2 (100%) Skin Sens. 1 (100%) Acute Tox. 3 (29.84%) STOT SE 3 (100%) Muta. 1B (100%) Carc. 2 (22.58%) Repr. 1B (77.42%) Repr. 1B (22.58%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H312 (29.84%): Harmful in contact with skin [Warning Acute toxicity, dermal] H315 (100%): Causes skin irritation [Warning Skin corrosion/irritation] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H331 (29.84%): Toxic if inhaled [Danger Acute toxicity, inhalation] H335 (100%): May cause respiratory irritation [Warning Specific target organ toxicity, single exposure; Respiratory tract irritation] H340 (100%): May cause genetic defects [Danger Germ cell mutagenicity] H351 (22.58%): Suspected of causing cancer [Warning Carcinogenicity] H360 (77.42%): May damage fertility or the unborn child [Danger Reproductive toxicity] H360FD (22.58%): May damage fertility; May damage the unborn child [Danger Reproductive toxicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Skin Irrit. 2H315 Skin Sens. 1H317 STOT SE 3H335 Muta. 1BH340 Aquatic Acute 1 H400 Aquatic Chronic 1 H410 Repr. 1B H360FD
1820573-27-0	XX	Beta-cyfluthrin	1							acutely toxic to mammals & birds LD50<200mg/kg body weight, aquatic toxicity LC/EC5<50µg/l Semi-volatile Non-mobile in soils Potential for bioaccumulation Toxic to mammals Invertebrates, aquatic plants and honeybees Toxic to birds, algae and earthworms Reproductive / developmental toxicant irritant disturbances Highly toxic to fish, aquatic Neurotoxicant Respiratory tract May cause metabolic or neurological	Lact. 1 Acute Tox. 2 Acute Tox. 2 STOT SE 1 Aquatic Acute 1 Aquatic Chronic 1		H300: Fatal if swallowed [Danger Acute toxicity, oral] H330: Fatal if inhaled [Danger Acute toxicity, inhalation] H362: May cause harm to breast-fed children [Reproductive toxicity, effects on or via lactation] H370: Causes damage to organs [Danger Specific target organ toxicity, single exposure] H400: Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410: Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox.4 H330
	X	BHC (mixture of various isomers)						1	1	Marine & water pollutant POP Chemical subject to PIC regulations Persistent is soil Public exposure routes - May be absorbed through the skin Occupational exposure routes - May be absorbed through the skin Toxic to mammals Toxic to birds & aquatic organisms Possible neurotoxicant respiratory tract, skin & eye irritant Exposure may cause dyspnoea, anorexia, tremors, convulsions and cramps Liver, kidney and heart toxicant Induces a carcinogenic response in rodents via aberrant cell proliferation US NTP - suspected carcinogen human carcinogen Highly volatile - If applied directly to plants or soil, drift is a concern & mitigation is advisable Slightly mobile Toxic to Possible carcinogen Possible	Acute Tox. 3 (90%) Acute Tox. 4 (70%) Carc. 2 (70%) Aquatic Acute 1 (70%) Aquatic Chronic 1 (70%)		H301 (90%): Toxic if swallowed [Danger Acute toxicity, oral] H312 (70%): Harmful in contact with skin [Warning Acute toxicity, dermal] H351 (70%): Suspected of causing cancer [Warning Carcinogenicity] H400 (70%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (70%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 Acute Tox. 4 Aquatic Acute 1 H400 Carc. 2H351 Aquatic Chronic 1 H410

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82657-04-3	XX	Bifenthrin							Neurotoxicant Possible carcinogen reproductive and developmental toxicant Toxic to birds , earthworms and bees Toxic to aquatic organisms This pesticide is a PFAS based on the OECD definition	EDC Possible	Acute Tox. 2 (92.78%) Skin Sens. 1B (95.36%) Acute Tox. 3 (95.36%) Carc. 2 (93.81%) STOT RE 1 (100%) Aquatic Acute 1 (97.42%) Aquatic Chronic 1 (97.94%)		H300 (92.78%): Fatal if swallowed [Danger Acute toxicity, oral] H317 (95.36%): May cause an allergic skin reaction [Warning Sensitization, Skin] H331 (95.36%): Toxic if inhaled [Danger Acute toxicity, inhalation] H351 (93.81%): Suspected of causing cancer [Warning Carcinogenicity] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H400 (97.42%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (97.94%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2H300 Skin Sens. 1BH317 Acute Tox. 3H331 Carc. 2H351 STOT RE 1H372 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
485-31-4	XXX	Binapacryl					1		Severe Marine Pollutant (Class O) - subject to PIC regulations Persistent in soil to aquatic organisms May cause reproductive and developmental effects Shows teratogenicity	Rotterdam Convention Highly toxic Toxic to birds and bees Possible neurotoxicant	Acute Tox. 4 (100%) Acute Tox. 4 (100%) Repr. 1B (59.42%) Repr. 1B (40.58%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H302+H312 (40.58%): Harmful if swallowed or in contact with skin [Warning Acute toxicity, oral; acute toxicity, dermal] H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H312 (100%): Harmful in contact with skin [Warning Acute toxicity, dermal] H360 (59.42%): May damage fertility or the unborn child [Danger Reproductive toxicity] H360D (40.58%): May damage the unborn child [Danger Reproductive toxicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 H302 Acute Tox. 4 H312 Aquatic Acute 1 H400 Aquatic Chronic 1 H410 Repr. 1B H360D
		Borax						1	Moderately toxic to earthworms aquatic organisms Possible neurotoxicant tract, skin and eye irritant	Moderately toxic to Possible EDC Respiratory	Eye Irrit. 2 (12.02%) Repr. 1B (94.41%)		H319 (12.02%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H360 (94.41%): May damage fertility or the unborn child [Danger Reproductive toxicity]	Repr. 1B H360
10043-35-3	XX	Boric acid					1		Toxic to aquatic environment and eye irritant	Respiratory tract, skin and eye irritant	Repr. 1B (100%)		H360 (100%): May damage fertility or the unborn child [Danger Reproductive toxicity]	Repr. 1B H360
		Boron						1	Toxic to aquatic environment May cause nausea, vomiting & diarrhoea if ingested	Skin irritant	Acute Tox. 4 (96.93%)		H302 (96.93%): Harmful if swallowed [Warning Acute toxicity, oral]	Acute Tox. 4H302 Aquatic Chronic 4 Skin Irrit. 2 Flam. Sol. 1 Acute Tox. 4 Pyr. Sol. 1 Aquatic Chronic 3
56073-10-0	XX	Brodifacoum					1		Semi-volatile soils & water 2nd generation coagulant Severe Marine Pollutant Persistent in soil and moderately persistent in water Acutely toxic to mammals & birds LD50<200mg/kg body weight Highly toxic to aquatic organisms LC/EC5<50µg/l poisonings may lead to massive haemorrhaging	Moderately persistent in Severe Marine Persistent in soil and moderately persistent in water Aquatic toxicity Highly toxic and severe poisonings may lead to massive haemorrhaging	Acute Tox. 1 (100%) Acute Tox. 1 (100%) Skin Sens. 1 (12.55%) Acute Tox. 1 (81.39%) Repr. 1A (68.83%) Repr. 1A (12.12%) STOT RE 1 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300+H310+H330 (28.14%): Fatal if swallowed, in contact with skin or if inhaled [Danger Acute toxicity, oral; acute toxicity, dermal; acute toxicity, inhalation] H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H310 (100%): Fatal in contact with skin [Danger Acute toxicity, dermal] H317 (12.55%): May cause an allergic skin reaction [Warning Sensitization, Skin] H330 (81.39%): Fatal if inhaled [Danger Acute toxicity, inhalation] H360 (68.83%): May damage fertility or the unborn child [Danger Reproductive toxicity] H360D (12.12%): May damage the unborn child [Danger Reproductive toxicity] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 1 H300 Repr. 1A H360D STOT RE 1 H372 STOT RE 2 H373 Acute Tox. 1H310 Acute Tox. 1H330 STOT RE 1H372 (blood) Aquatic Acute 1H400 Aquatic Chronic 1 H410 Repr. 1AH360D

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28772-56-7	XX	Bromadiolone	1	1						Second generation anticoagulant persistent in soil and water Reproductive and developmental toxin Possible EDC Very toxic by inhalation, contact with the skin and by ingestion acutely toxic to mammals & birds LD50<200mg/kg body weight Highly toxic to mammals	Moderately	Acute Tox. 1 (79.12%) Acute Tox. 4 (20.88%) Acute Tox. 1 (78.71%) Acute Tox. 1 (63.45%) Repr. 1B (50.6%) Repr. 1B (11.24%) STOT RE 1 (63.86%) Aquatic Acute 1 (78.31%) Aquatic Chronic 1 (63.86%)		H300+H310+H330 (26.1%): Fatal if swallowed, in contact with skin or if inhaled [Danger Acute toxicity, oral; acute toxicity, dermal; acute toxicity, inhalation] H300 (79.12%): Fatal if swallowed [Danger Acute toxicity, oral] H302 (20.88%): Harmful if swallowed [Warning Acute toxicity, oral] H310 (78.71%): Fatal in contact with skin [Danger Acute toxicity, dermal] H330 (63.45%): Fatal if inhaled [Danger Acute toxicity, inhalation] H360 (50.6%): May damage fertility or the unborn child [Danger Reproductive toxicity] H360D (11.24%): May damage the unborn child [Danger Reproductive toxicity] H372 (63.86%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H400 (78.31%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (63.86%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 1H300 Repr. 1B H360D STOT RE 1 H372 Acute Tox. 1 H310 Acute Tox. 1H372 (blood) Aquatic Acute 1H400 Aquatic Chronic 1 H410 Repr. 1BH360D
1689-84-5	XX	Bromoxynil								Marine Pollutant High aqueous solubility Volatile Low volatility. If applied directly to plants, drift is a concern & mitigation is advisable Non-persistent in water Moderately mobile in soils Low leachability in soils Substance of concern for long-range air transport toxic to mammals Highly Toxic to mammals, birds, earthworms & honeybees Acutely toxic to mammals & birds LD50<200mg/kg body weight Toxic to aquatic organisms Aquatic toxicity LC/EC5<50µg/l Reproductive/ developmental toxicant Skin sensitiser Possible carcinogen		Acute Tox. 3 (100%) Skin Sens. 1 (100%) Acute Tox. 2 (100%) Repr. 2 (75.2%) Repr. 2 (24.8%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H330 (100%): Fatal if inhaled [Danger Acute toxicity, inhalation] H361 (75.2%): Suspected of damaging fertility or the unborn child [Warning Reproductive toxicity] H361d (24.8%): Suspected of damaging the unborn child [Warning Reproductive toxicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 H301 Skin Sens. 1 H317 Acute Tox. 2 H330 Aquatic Acute 1 H400 Aquatic Chronic 1 H410 Repr. 2 H361d
		Bromoxynil octanoate								Toxic to the aquatic environment reproductive and developmental toxicant Skin sensitiser	Possible	Acute Tox. 4 (69.29%) Acute Tox. 3 (30.71%) Repr. 2 (80%) Repr. 2 (21.43%) Acute Tox. 4 (56.43%) Skin Sens. 1 (100%) Acute Tox. 3 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H301+H331 (20%): Toxic if swallowed or if inhaled [Danger Acute toxicity, oral; acute toxicity, inhalation] H301 (30.71%): Toxic if swallowed [Danger Acute toxicity, oral] H302+H312 (27.14%): Harmful if swallowed or in contact with skin [Warning Acute toxicity, oral; acute toxicity, dermal] H302 (69.29%): Harmful if swallowed [Warning Acute toxicity, oral] H312 (56.43%): Harmful in contact with skin [Warning Acute toxicity, dermal] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H331 (100%): Toxic if inhaled [Danger Acute toxicity, inhalation] H361 (80%): Suspected of damaging fertility or the unborn child [Warning Reproductive toxicity] H361d (21.43%): Suspected of damaging the unborn child [Warning Reproductive toxicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 H302 Skin Sens. 1 H317 Acute Tox. 3 H331 Aquatic Acute 1 H400 Aquatic Chronic 1 H410 Repr. 2 H361d
	Y	C9 Solvent naphtha	1	1						Potential marine pollutant Reproductive / developmental toxicant & skin & eye irritant Possible neurotoxicant Harmful if swallowed or inhaled Skin contact may lead to drying or dermatitis Possible liver and kidney toxicant genetic defects	Carcinogen Respiratory tract May cause				

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95465-99-9	XX	Cadusafos	1						1			Highly toxic to mammals	Acute Tox. 2 (100%) Acute Tox. 1 (100%) Acute Tox. 1 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (29.63%)		H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H310 (100%): Fatal in contact with skin [Danger Acute toxicity, dermal] H330 (100%): Fatal if inhaled [Danger Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (29.63%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 Acute Tox. 1 Acute Tox. 1 Aquatic Acute 1 Acute Tox. 2 Aquatic Chronic 1
50-14-6	XX	Calciferol							1			Highly toxic to mammals	Acute Tox. 2 (37.83%) Acute Tox. 3 (37.83%) Acute Tox. 2 (61.74%) Acute Tox. 2 (37.83%) Acute Tox. 3 (61.3%) Acute Tox. 2 (99.13%) STOT RE 1 (100%) Aquatic Chronic 4 (37.83%)		H300+H310+H330 (37.83%): Fatal if swallowed, in contact with skin or if inhaled [Danger Acute toxicity, oral; acute toxicity, dermal; acute toxicity, inhalation] H300 (37.83%): Fatal if swallowed [Danger Acute toxicity, oral] H301 (61.74%): Toxic if swallowed [Danger Acute toxicity, oral] H310 (37.83%): Fatal in contact with skin [Danger Acute toxicity, dermal] H311 (61.3%): Toxic in contact with skin [Danger Acute toxicity, dermal] H330 (99.13%): Fatal if inhaled [Danger Acute toxicity, inhalation] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H413 (37.83%): May cause long lasting harmful effects to aquatic life [Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 H301 Acute Tox. 3 H311 Acute Tox. 2 H330 STOT RE 1 H372
		Calcium arsenate	1	1					1		Neurotoxicant Very persistent in soils Toxic to aquatic organisms mammals health effects Liver & kidney toxicant May cause marrow damage	Environmental toxicant Toxic to bees Very toxic to Highly toxic - causes systemic	Acute Tox. 3 (100%) Acute Tox. 3 (100%) Carc. 1A (100%) Aquatic Acute 1 (96.3%) Aquatic Chronic 1 (100%)		H301+H331 (81.48%): Toxic if swallowed or if inhaled [Danger Acute toxicity, oral; acute toxicity, inhalation] H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H331 (100%): Toxic if inhaled [Danger Acute toxicity, inhalation] H350 (100%): May cause cancer [Danger Carcinogenicity] H400 (96.3%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 H301 Acute Tox. 3 H331 Carc. 1A H350 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
8001-35-2	XXX	Camphechlor (CLC)							1		Very toxic to aquatic organisms	Possible carcinogen	Acute Tox. 3 (100%) Acute Tox. 4 (100%) Skin Irrit. 2 (100%) STOT SE 3 (100%) Carc. 2 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H312 (100%): Harmful in contact with skin [Warning Acute toxicity, dermal] H315 (100%): Causes skin irritation [Warning Skin corrosion/irritation] H335 (100%): May cause respiratory irritation [Warning Specific target organ toxicity, single exposure; Respiratory tract irritation] H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 H301 Acute Tox. 4 H312 Skin Irrit. 2 H315 STOT SE 3 H335 Carc. 2 H351 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
6/1/2425	XXX	Captafol	1	1					1		Rotterdam Convention (Class Ia) - subject to PIC regulations Toxic to bees Toxic to birds and earthworms Very toxic to aquatic organisms Occupational: Contact dermatitis, severe irritation of the respiratory tract, eye damage		Skin Sens. 1 (100%) Carc. 1B (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H350 (100%): May cause cancer [Danger Carcinogenicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Skin Sens. 1 H317 Carc. 1B H350 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
		Captan							1		Toxic to mammals, birds & earthworms Toxic to aquatic organisms EDC Irritant	Toxic Possible carcinogen Skin & eye May cause contact dermatitis	Skin Sens. 1 (100%) Eye Dam. 1 (100%) Acute Tox. 3 (100%) Carc. 2 (100%) Aquatic Acute 1 (100%)		H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H318 (100%): Causes serious eye damage [Danger Serious eye damage/eye irritation] H331 (100%): Toxic if inhaled [Danger Acute toxicity, inhalation] H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard]	Eye Dam. 1 H318 Skin Sens. 1 H317 Acute Tox. 3 H331 Carc. 2 H351 Aquatic Acute 1 H400

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		Carbaryl							Obsolete carbamate insecticide subject to PIC regulations - CANDIDATE FOR LISTING Marine Pollutant Highly toxic to mammals Toxic to aquatic organisms Possible carcinogen developmental toxicant Aquatic toxicity LC/EC5<=50µg/l	Chemical Volatile Toxic to birds EDC Reproductive and Acetyl cholinesterase inhibitor	Acute Tox. 3 (64.42%) Acute Tox. 4 (35.58%) Acute Tox. 4 (97.12%) Carc. 2 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (70.19%)		H301 (64.42%): Toxic if swallowed [Danger Acute toxicity, oral] H302 (35.58%): Harmful if swallowed [Warning Acute toxicity, oral] H332 (97.12%): Harmful if inhaled [Warning Acute toxicity, inhalation] H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (70.19%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 H302 Acute Tox. 4 H332 Carc. 2 H351 Aquatic Acute 1 H400
63-25-2	XX	Carbendazim					1	1	Marine Pollutant mobile Moderately persistent in soil & water Toxic to honeybees and most aquatic organisms Highly toxic to earthworms Public exposure routes - Negligible risk to consumers and bystanders Low risk to agricultural workers Occupational exposure routes - Negligible risk to consumers and bystanders Genotoxic - May cause genetic defects EDC - Increase of estrogen production and aromatase activity Reproductive / developmental toxicant Possible carcinogen Evidence of liver enzyme induction Possible liver toxicant US EPA - possible human carcinogen	Volatile and moderately mobile Moderately persistent in soil & water Toxic to honeybees Highly toxic to earthworms Public exposure routes - Negligible risk to consumers and bystanders Low risk to agricultural workers Occupational exposure routes - Negligible risk to consumers and bystanders Genotoxic - May cause genetic defects EDC - Increase of estrogen production and aromatase activity Reproductive / developmental toxicant Possible carcinogen Evidence of liver enzyme induction Possible liver toxicant US EPA - possible human carcinogen	Skin Sens. 1 (21.2%) Muta. 1B (99.86%) Repr. 1B (83.54%) Repr. 1B (18.97%) Aquatic Acute 1 (98.61%) Aquatic Chronic 1 (100%)		H317 (21.2%): May cause an allergic skin reaction [Warning Sensitization, Skin] H340 (99.86%): May cause genetic defects [Danger Germ cell mutagenicity] H360 (83.54%): May damage fertility or the unborn child [Danger Reproductive toxicity] H360FD (18.97%): May damage fertility; May damage the unborn child [Danger Reproductive toxicity] H400 (98.61%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Skin Sens. 1H317 Muta. 1BH340 Aquatic Acute 1, H400 Aquatic Chronic 1 H410 Repr. 1B H360
10605-21-7	XX	Carbendazim + flusilazole				1	1	1	Marine Pollutant Volatile & moderately mobile in soils persistent in soil & water Highly toxic to earthworms organisms Public exposure routes - Negligible risk to consumers and bystanders Low risk to agricultural workers Occupational exposure routes - Negligible risk to consumers and bystanders Genotoxic - May cause genetic defects EDC - Increase of estrogen production and aromatase activity Genotoxic - Chromosome aberration (EFSA), Gene mutation (EFSA) & Unspecified genotoxicity type sensitiser Reproductive / developmental toxicant Possible carcinogen Evidence of liver enzyme induction Possible liver toxicant US EPA - possible human carcinogen	Low water solubility Moderately mobile Toxic to honeybees Highly toxic to aquatic organisms Public exposure routes - Negligible risk to consumers and bystanders Low risk to agricultural workers Occupational exposure routes - Negligible risk to consumers and bystanders Genotoxic - May cause genetic defects EDC - Increase of estrogen production and aromatase activity Genotoxic - Chromosome aberration (EFSA), Gene mutation (EFSA) & Unspecified genotoxicity type sensitiser Reproductive / developmental toxicant Possible carcinogen Evidence of liver enzyme induction Possible liver toxicant US EPA - possible human carcinogen	Skin Sens. 1 (21.2%) Muta. 1B (99.86%) Repr. 1B (83.54%) Repr. 1B (18.97%) Aquatic Acute 1 (98.61%) Aquatic Chronic 1 (100%) Acute Tox. 4 (100%) Carc. 2 (100%) Aquatic Chronic 2 (100%)		H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H317 (21.2%): May cause an allergic skin reaction [Warning Sensitization, Skin] H340 (99.86%): May cause genetic defects [Danger Germ cell mutagenicity] H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H360 (83.54%): May damage fertility or the unborn child [Danger Reproductive toxicity] H360D (34.15%): May damage the unborn child [Danger Reproductive toxicity] H360FD (18.97%): May damage fertility; May damage the unborn child [Danger Reproductive toxicity] H400 (98.61%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard] H411 (100%): Toxic to aquatic life with long lasting effects [Hazardous to the aquatic environment, long-term hazard]	Skin Sens. 1H317 Muta. 1BH340 Aquatic Acute 1 H400 Aquatic Chronic 1 H410 Repr. 1B H360D Acute Tox. 4 H302 Carc. 2 H351 Aquatic Chronic 2 H411 Repr. 1B H360FD
1563-66-2	XX	Carbofuran				1	1	1	Potential groundwater pollutant Persistent in water Highly toxic to mammals Toxic to earthworms aquatic organisms Potential Acetyl cholinesterase inhibitor Potential neurotoxicant issues - Increase of progesterone, cortisol and estradiol levels Skin sensitiser and possible skin irritant Possible reproductive and developmental toxicant Possible genotoxic (chromosome aberration) inhibitor degeneration May be fatal if swallowed, inhaled, or absorbed through skin	Potential marine pollutant Chemical subject to PIC regulations - Bioaccumulates Highly toxic to birds Very toxic to aquatic organisms EDC - Endocrine inhibitor Acetyl cholinesterase inhibitor May cause testicular	Acute Tox. 1 (100%) Acute Tox. 2 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300+H330 (21.88%): Fatal if swallowed or if inhaled [Danger Acute toxicity, oral; acute toxicity, inhalation] H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H330 (100%): Fatal if inhaled [Danger Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 H300 Acute Tox. 2 H330 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
55285-14-8	XX	Carbosulfan						1	Chemical subject to PIC regulations -RECOMMENDED FOR LISTING Toxic to mammals Skin sensitiser and possible skin irritant Environmental toxicant Bioaccumulates Possible genotoxic (chromosome aberration) Acetyl cholinesterase inhibitor Possible reproductive and developmental toxicant		Acute Tox. 3 (100%) Skin Sens. 1 (100%) Acute Tox. 2 (96.43%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H330 (96.43%): Fatal if inhaled [Danger Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 H301 Skin Sens. 1 H317 Acute Tox. 2 H330 Aquatic Acute 1 H400 Aquatic Chronic 1 H410

UnPoison South Africa's Highly Hazardous Pesticide Database vs EU's According to JMPM Criteria (FAO/WHO), GHS and ECHA Classification Systems - February 2025

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UnPoison South Africa's Highly Hazardous Pesticide Database vs EU's According to JMPM Criteria (FAO/WHO), GHS and ECHA Classification Systems - February 2025

76-062	XX	Chloropicrin								1	Volatile Persistent in water concern for long-range air transport Bioaccumulation potential honeybees, bumble bees and earthworms Highly toxic to aquatic organisms EDC - Endocrine issues - Activation of androgen-sensitive cells proliferation developmental toxicant Respiratory tract, skin & eye irritant Skin sensitiser carcinogen dermatitis Slightly mobile Substance of Toxic to birds, Possible May cause contact Acute Tox. 3 (23.38%) Acute Tox. 4 (76.62%) Acute Tox. 2 (23.38%) Skin Irrit. 2 (100%) Eye Irrit. 2 (100%) Acute Tox. 2 (100%) STOT SE 3 (100%) Aquatic Acute 1 (23.38%) Aquatic Chronic 1 (23.38%)		H301 (23.38%): Toxic if swallowed [Danger Acute toxicity, oral] H302 (76.62%): Harmful if swallowed [Warning Acute toxicity, oral] H310+H330 (23.38%): Fatal in contact with skin or if inhaled [Danger Acute toxicity, dermal; acute toxicity, inhalation] H310 (23.38%): Fatal in contact with skin [Danger Acute toxicity, dermal] H315 (100%): Causes skin irritation [Warning Skin corrosion/irritation] H319 (100%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H330 (100%): Fatal if inhaled [Danger Acute toxicity, inhalation] H335 (100%): May cause respiratory irritation [Warning Specific target organ toxicity, single exposure; Respiratory tract irritation] H400 (23.38%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (23.38%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 Skin Irrit. 2 Eye Irrit. 2 H319 Acute Tox. 2 STOT SE 3 H330 H335
1897-45-6	XX	Chlorothalonil								1	Persistent-Bioaccumulative-Toxic' Pollutant contaminant Non-volatile biodegradable Toxic to birds & earthworms organisms Reproductive / developmental toxicant Marine Potential groundwater Moderate solubility in water Not readily Moderately mobile in soils Toxic to aquatic Possible carcinogen, EDC, Skin Sens. 1 (100%) Eye Dam. 1 (100%) Acute Tox. 2 (97.27%) STOT SE 3 (100%) Carc. 2 (97.27%) Aquatic Acute 1 (96.97%) Aquatic Chronic 1 (100%)		H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H318 (100%): Causes serious eye damage [Danger Serious eye damage/eye irritation] H330 (97.27%): Fatal if inhaled [Danger Acute toxicity, inhalation] H335 (100%): May cause respiratory irritation [Warning Specific target organ toxicity, single exposure; Respiratory tract irritation] H351 (97.27%): Suspected of causing cancer [Warning Carcinogenicity] H400 (96.97%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Eye Dam. 1H318 Skin Sens. 1H317 Acute Tox. 2 H330 STOT SE 3H335 Carc. 2H351 Aquatic Acute 1H400 Aquatic Chronic 1 H410
		Chlorotoluron								1	Moderately soluble in water Highly volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable Moderately mobile May bioaccumulate honeybees Possible carcinogen, Acetyl cholinesterase inhibitor, neurotoxicant, reproductive / developmental toxicant Possible spleen, bone-marrow & red blood cell toxicant Highly volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable Persistent in water Toxic to birds & honeybees Toxic to aquatic organisms Possible carcinogen, Acetyl cholinesterase inhibitor, neurotoxicant, reproductive / developmental toxicant Possible spleen, bone-marrow & red blood cell toxicant Carc. 2 (100%) STOT RE 2 (89.4%) Repr. 2 (89.4%) Repr. 2 (13.47%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H361 (89.4%): Suspected of damaging fertility or the unborn child [Warning Reproductive toxicity] H361d (13.47%): Suspected of damaging the unborn child [Warning Reproductive toxicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Carc. 2 H351 Aquatic Acute 1 H400 Aquatic Chronic 1 H410 Repr. 2 H361d
101-21-3	XX	Chlorpropham								1	PIC candidate - DGD's under development Marine Pollutant solubility Non-mobile to groundwater Highly toxic to birds, fish, aquatic invertebrates & honeybees Toxic to aquatic plants, algae & earthworms - threshold for concern Occupational exposure routes - PPE/PPC required to mitigate any exposure risks EDC developmental toxicant Neurotoxicant Skin and eye irritant ingestion Some studies indicate a link between chlorpyrifos metabolites and learning difficulties in children Suspected cardiovascular & blood toxicant Low aqueous Quite volatile Risk of leaching to groundwater Persistent in soils & water Bioconcentration Public exposure routes - Possible Highly toxic to mammals Reproductive / Acetyl cholinesterase inhibitor Skin sensitiser Highly toxic by ingestion Carc. 2 (100%) STOT RE 2 (100%) Aquatic Chronic 2 (100%)		H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H373 (100%): May causes damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H411 (100%): Toxic to aquatic life with long lasting effects [Hazardous to the aquatic environment, long-term hazard]	Carc. 2 H351 STOT RE 2 H373 Aquatic Chronic 2 H411

		Chlorpyrifos								Rotterdam Convention - subject to PIC regulations- CHEMICALS RECOMMENDED FOR LISTING Water pollutant Potential groundwater contaminant Marine pollutant Strong oxidizer Not mobile in soils Persistent in soils and water Toxic to aquatic organisms Skin sensitizer Carcinogenic Genotoxic Possible reproductive / developmental toxicant Respiratory tract irritant	Acute Tox. 3 (100%) Acute Tox. 4 (13.7%) Eye Irrit. 2 (13.7%) Acute Tox. 1 (15.93%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H312 (13.7%): Harmful in contact with skin [Warning Acute toxicity, dermal] H319 (13.7%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H330 (15.93%): Fatal if inhaled [Danger Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 H312 Aquatic Acute 1 H400 Aquatic Chronic 1 H410	
2921-88-2	XXX								1						
		Chromium trioxide								Obsolete first generation anticoagulant rodenticide Low aqueous solubility Non Volatile Moderately persistent in soil systems and aquatic systems Moderately mobile Risk to leach to groundwater Bioaccumulation a concern Toxic to birds, fish and aquatic invertebrates Highly toxic to mammals with high oral toxicity Possible respiratory tract & skin irritant Highly toxic Anticoagulant - may cause a gastrointestinal hemorrhage	Ox. Sol. 1 (100%) Acute Tox. 3 (90.95%) Acute Tox. 2 (43.78%) Acute Tox. 3 (47.82%) Skin Corr. 1A (100%) Skin Sens. 1 (100%) Eye Dam. 1 (18.26%) Acute Tox. 2 (90.95%) Resp. Sens. 1 (97.42%) Muta. 1B (99.35%) Carc. 1A (100%) Repr. 2 (85.46%) Repr. 2 (14.54%) STOT RE 1 (100%) Aquatic Acute 1 (89.98%) Aquatic Chronic 1 (91.44%)		H271 (100%): May cause fire or explosion; strong Oxidizer [Danger Oxidizing liquids; Oxidizing solids] H301 (90.95%): Toxic if swallowed [Danger Acute toxicity, oral] H310 (43.78%): Fatal in contact with skin [Danger Acute toxicity, dermal] H311 (47.82%): Toxic in contact with skin [Danger Acute toxicity, dermal] H314 (100%): Causes severe skin burns and eye damage [Danger Skin corrosion/irritation] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H318 (18.26%): Causes serious eye damage [Danger Serious eye damage/eye irritation] H330 (90.95%): Fatal if inhaled [Danger Acute toxicity, inhalation] H334 (97.42%): May cause allergy or asthma symptoms or breathing difficulties if inhaled [Danger Sensitization, respiratory] H340 (99.35%): May cause genetic defects [Danger Germ cell mutagenicity] H350 (100%): May cause cancer [Danger Carcinogenicity] H361 (85.46%): Suspected of damaging fertility or the unborn child [Warning Reproductive toxicity] H361f (14.54%): Suspected of damaging fertility [Warning Reproductive toxicity] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H400 (89.98%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (91.44%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Ox. Sol. 1H271 STOT SE 3 H335 Acute Tox. 3 H301 Acute Tox. 3 H311 Skin Corr. 1AH314 Skin Sens. 1H317 Acute Tox. 2 H300 Resp. Sens. 1H334 Muta. 1B H 340 Carc. 1A H350 STOT RE 1H373 Aquatic Acute 1H400 Aquatic Chronic 1 H410 Repr. 2 H361f	
			1	1											
		Coumatetralyl								Potential groundwater contaminant Highly soluble in water Highly volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable Very persistent in water Very mobile High potential for bio-concentration Substance of concern for long-range air transport Toxic to birds, fish, aquatic invertebrates, aquatic fauna, honeybees and earthworms Occupational exposure routes - Exposure possible via ingestion or inhalation Toxic to mammals Skin irritant Skin sensitiser Eye irritant Possible carcinogen Respiratory tract irritant & Reproductive / developmental toxicant Toxic if swallowed or in contact with skin Potential for vomiting, parasympathetic hyperactivity, dyspnea, hypotension and tachycardia	Acute Tox. 2 (100%) Acute Tox. 2 (97.67%) STOT RE 1 (100%) Aquatic Chronic 3 (97.67%)		H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H310 (97.67%): Fatal in contact with skin [Danger Acute toxicity, dermal] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H412 (97.67%): Harmful to aquatic life with long lasting effects [Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 STOT RE 1 Acute Tox. 3 Acute Tox. 2 Aquatic Chronic 1 Repr. 1B	
5836-29-3	XX		1						1						

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420-04-2	XX	Cyanamide								1	Low aqueous solubility Moderately persistent in soils & water Bio-concentration factor: threshold for concern Highly toxic to mammals, honey bees, bumble bees and other bees - Megachile rotundata & Nomia melanderi Highly toxic to aquatic organisms Skin irritant developmental toxicant Possible liver or kidney toxicant Non-volatile Non-mobile Neurotoxicant Possible reproductive / Highly toxic Water-react. 3 (23.53%) Acute Tox. 4 (100%) Skin Irrit. 2 (52.94%) Skin Sens. 1 (74.12%) Eye Dam. 1 (100%) STOT SE 3 (100%) Aquatic Chronic 3 (51.18%)		H261 (23.53%): In contact with water releases flammable gas [Danger Substances and mixtures which in contact with water, emit flammable gases] H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H315 (52.94%): Causes skin irritation [Warning Skin corrosion/irritation] H317 (74.12%): May cause an allergic skin reaction [Warning Sensitization, Skin] H318 (100%): Causes serious eye damage [Danger Serious eye damage/eye irritation] H335 (100%): May cause respiratory irritation [Warning Specific target organ toxicity, single exposure; Respiratory tract irritation] H412 (51.18%): Harmful to aquatic life with long lasting effects [Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 Acute Tox. 3 Skin Corr. 1 Eye Dam. 1 Skin Sens. 1 Carc. 2 STOT RE 2 Aquatic Chronic 3 Repr. 2
68359-37-5	XX	Cyfluthrin	1							1	Possible groundwater pollutant subject to PIC regulations Not mobile & low leachability Bioaccumulates mammals Toxic to aquatic organisms Skin & eye irritant Chemicals Persistent in soil and water Toxic to Very toxic to honey bees EDC Acute Tox. 2 (100%) Acute Tox. 2 (84.09%) Acute Tox. 3 (17.8%) Lact. (43.56%) STOT SE 1 (43.56%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H330 (84.09%): Fatal if inhaled [Danger Acute toxicity, inhalation] H331 (17.8%): Toxic if inhaled [Danger Acute toxicity, inhalation] H362 (43.56%): May cause harm to breast-fed children [Reproductive toxicity, effects on or via lactation] H370 (43.56%): Causes damage to organs [Danger Specific target organ toxicity, single exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 Acute Tox. 2 Lact. STOT SE 1 Aquatic Acute 1 Aquatic Chronic 1
94361-06-5	XX	Cyproconazole								1	Marine Pollutant Relatively volatile to groundwater Non-mobile concentration factor - threshold for concern Toxic to mammals Toxic to aquatic organisms humans Carcinogen EDC - Endocrine issues - Competitive binding to androgen receptors Neurotoxicant developmental toxicant Highly toxic by ingestion, inhalation and via skin absorption Strong links with breast and womb cancer acutely toxic to mammals & birds LD50-200mg/kg body weight Low aqueous solubility Low potential to leach to groundwater Bio- Highly toxic to honey bees & other bee species - Megachile rotundata & Nomia melanderi Toxic to aquatic organisms Toxic by the oral route in humans Reproductive / Unspecified genotoxicity type Possible respiratory tract irritant Acute Tox. 3 (54.08%) Acute Tox. 4 (45.92%) Repr. 1B (29%) Repr. 1B (26.89%) Repr. 2 (45.62%) STOT RE 2 (54.08%) Aquatic Acute 1 (99.7%) Aquatic Chronic 1 (99.7%)		H301 (54.08%): Toxic if swallowed [Danger Acute toxicity, oral] H302 (45.92%): Harmful if swallowed [Warning Acute toxicity, oral] H360 (29%): May damage fertility or the unborn child [Danger Reproductive toxicity] H360D (26.89%): May damage the unborn child [Danger Reproductive toxicity] H361 (45.62%): Suspected of damaging fertility or the unborn child [Warning Reproductive toxicity] H373 (54.08%): May causes damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H400 (99.7%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (99.7%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 STOT RE 2 Aquatic Acute 1 Aquatic Chronic 1 Repr. 1B
	XX	DDT								1	Marine pollutant aqueous solubility Low potential to leach to groundwater soil Bio-concentration factor - threshold for concern Highly toxic to most aquatic organisms & honeybees, mason bees, bumble bees & other bee species - Megachile rotundata & Nomia melanderi Slight risk of dietary exposure for small children however ARfD would not normally be exceeded Neurotoxin Weak estrogenic activity Low Semi-volatile Not persistent in Non-mobile Public exposure routes - Slight Highly toxic to mammals EDC - Endocrine issues - Acute Tox. 3 (98.85%) Acute Tox. 3 (50.57%) Carc. 2 (98.85%) STOT RE 1 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (98.85%)		H301+H311 (43.68%): Toxic if swallowed or in contact with skin [Danger Acute toxicity, oral; acute toxicity, dermal] H301 (98.85%): Toxic if swallowed [Danger Acute toxicity, oral] H311 (50.57%): Toxic in contact with skin [Danger Acute toxicity, dermal] H351 (98.85%): Suspected of causing cancer [Warning Carcinogenicity] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (98.85%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 Carc. 2 STOT RE 1 Aquatic Acute 1 Aquatic Chronic 1

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62-73-7	XX	Dichlorvos	1							Severe Marine Pollutant solubility groundwater Non-persistent in soils but persistent in aquatic systems Non-mobile in soils Toxic to birds, most aquatic organisms & earthworms Toxic to mammals if ingested Slight risk to the public from treated recreational areas such as golf courses irritant Kidney and liver toxicant May cause bronchopneumonia or pulmonary oedema	Low aqueous solubility Low risk of leaching to groundwater Relatively volatile Toxic to birds, most aquatic organisms & earthworms Public exposure routes - Slight risk to the public from treated recreational areas such as golf courses Respiratory tract irritant Skin sensitiser May cause bronchopneumonia or pulmonary oedema	Acute Tox. 2 (41.73%) Acute Tox. 3 (58.27%) Acute Tox. 3 (94.24%) Skin Sens. 1 (100%) Acute Tox. 2 (62.59%) Acute Tox. 3 (37.41%) Aquatic Acute 1 (100%)		H300 (41.73%): Fatal if swallowed [Danger Acute toxicity, oral] H301 (58.27%): Toxic if swallowed [Danger Acute toxicity, oral] H311 (94.24%): Toxic in contact with skin [Danger Acute toxicity, dermal] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H330 (62.59%): Fatal if inhaled [Danger Acute toxicity, inhalation] H331 (37.41%): Toxic if inhaled [Danger Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard]	Acute Tox. 3 Acute Tox. 3 Skin Sens. 1 Acute Tox. 2 Aquatic Acute 1
		Diclofop-methyl								Severe Marine Pollutant Volatile Non-mobile in soils particle bound transport aquatic systems Toxic to birds, earthworms & honeybees (Ladybirds) - <i>Coccinella septempunctata</i> , Beneficial insects (Parasitic wasps) - <i>Phidius rhopalosiphi</i> , Beneficial insects (Predatory mites) - <i>Typhlodromus pyri</i> & Beneficial insects (Ground beetles) - <i>Poecilus cupreus</i> Public Exposure Routes - Slight risk to the public from treated recreational areas such as golf courses Respiratory tract irritant Possible carcinogen, reproductive / developmental toxicant & skin irritant May cause bronchopneumonia or pulmonary oedema US EPA - likely to be human carcinogen	Low aqueous solubility Non-persistent in soils Medium potential for persistent in aquatic systems Toxic to mammals if ingested Toxic to Beneficial insects Toxic to aquatic organisms Slight risk to the public from treated recreational areas such as golf courses Skin sensitiser Possible carcinogen, reproductive / developmental toxicant & skin irritant Kidney and liver toxicant May cause bronchopneumonia or pulmonary oedema	Acute Tox. 4 (100%) Skin Sens. 1 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 *H302 Skin Sens. 1H317 Aquatic Acute 1 Aquatic Chronic 1
7173-51-5	XX	Didecylmethylammonium chloride (DDAC)								BIOCIDE in water Non-mobile in soils aquatic invertebrates Highly toxic to aquatic invertebrates Toxic to mammals & Neurotoxicant	Volatility Highly toxic to fish Respiratory tract, skin & eye irritant Possible Reproductive / developmental toxicant May cause serious eye damage	Acute Tox. 3 (16%) Acute Tox. 4 (85.1%) Skin Corr. 1B (100%) Eye Dam. 1 (27.3%) Aquatic Acute 1 (46.9%) Aquatic Chronic 1 (21.3%) Aquatic Chronic 2 (30%)		H301 (16%): Toxic if swallowed [Danger Acute toxicity, oral] H302 (85.1%): Harmful if swallowed [Warning Acute toxicity, oral] H314 (100%): Causes severe skin burns and eye damage [Danger Skin corrosion/irritation] H318 (27.3%): Causes serious eye damage [Danger Serious eye damage/eye irritation] H400 (46.9%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (21.3%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard] H411 (30%): Toxic to aquatic life with long lasting effects [Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 H302 Skin Corr. 1BH314
60-57-1	XXX	Dieldrin								Persistent-Bioaccumulative-Toxic' Low aqueous solubility Slightly volatile Persistent in soil & aquatic environments bioaccumulation Toxic to mammals, birds, earthworms & most aquatic organisms Genotoxic - Chromosome aberration (EFSA) Potential carcinogen reproductive / developmental toxicant Liver, heart, thyroid and kidney toxicant	Mobile in soils Potential for particle bound transport Potential for bioaccumulation Toxic to mammals, birds, earthworms & most aquatic organisms Skin & eye irritant Potential	Acute Tox. 2 (15.79%) Acute Tox. 3 (84.21%) Acute Tox. 1 (100%) Acute Tox. 1 (15.79%) Carc. 2 (100%) STOT RE 1 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300 (15.79%): Fatal if swallowed [Danger Acute toxicity, oral] H301 (84.21%): Toxic if swallowed [Danger Acute toxicity, oral] H310 (100%): Fatal in contact with skin [Danger Acute toxicity, dermal] H330 (15.79%): Fatal if inhaled [Danger Acute toxicity, inhalation] H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 Acute Tox. 1 Carc. 2 STOT RE 1 Aquatic Acute 1 Aquatic Chronic 1

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56073-07-5	XX	Difenacoum	1						1	Low solubility in water Low potential to leach to groundwater Highly persistent in soils High potential for bioaccumulation to birds and aquatic life mammals birds LD50<200mg/kg body weight Occupational exposure routes - May be absorbed through the skin Reproductive / developmental toxicant May cause severe haemorrhage leading to death Fatal if inhaled - H330 (GHS) Relatively volatile Non-mobile in soils Highly toxic to mammals & birds Acutely toxic to mammals & birds Acute Tox. 1 (100%) Acute Tox. 1 (19.5%) Acute Tox. 1 (19.5%) Repr. 1B (17.61%) STOT RE 1 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300+H310+H330 [17.61%]: Fatal if swallowed, in contact with skin or if inhaled [Danger Acute toxicity, oral; acute toxicity, dermal; acute toxicity, inhalation] H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H310 (19.5%): Fatal in contact with skin [Danger Acute toxicity, dermal] H330 (19.5%): Fatal if inhaled [Danger Acute toxicity, inhalation] H360D (17.61%): May damage the unborn child [Danger Reproductive toxicity] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 1 Repr. 1B STOT RE 1 Acute Tox. 1 Acute Tox. 1 STOT RE 1 Aquatic Acute 1 Aquatic Chronic 1 Repr. 1B
		Difenoconazole							1	Low aqueous solubility Very persistent in soil systems High potential for bioaccumulation birds and fish Highly toxic to mammals developmental toxicant May cause blood coagulation and haemorrhaging Vapour may cause dizziness or suffocation Low volatility Non-mobile in soils Highly toxic to birds and fish Reproductive / developmentally toxicant Highly toxic Acute Tox. 4 (100%) Eye Irrit. 2 (51.56%) Acute Tox. 4 (31.25%) Aquatic Acute 1 (96.88%) Aquatic Chronic 1 (96.88%)		H302+H332 (29.69%): Harmful if swallowed or if inhaled [Warning Acute toxicity, oral; acute toxicity, inhalation] H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H319 (51.56%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H332 (31.25%): Harmful if inhaled [Warning Acute toxicity, inhalation] H400 (96.88%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (96.88%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 Acute Tox. 4 Aquatic Acute 1 Aquatic Chronic 1
104653-34-1	XX	Difethialone	1						1	Potential groundwater contaminant Low groundwater leaching potential Non-persistent in soil Mobile in soils species (LC/EC5<50µg/l) Toxic to mammals & earthworms (LD50<200mg/kg body weight) Genotoxic - DNA damage/repair (EFSA), Gene mutation (EFSA) Acetyl cholinesterase inhibitor Possible EDC - Endocrine issues - Disruption of thyroid hormones action swallowed Some evidence of neurotoxicity but not conclusive Fatal if inhaled - H330 (GHS) Severe Marine Pollutant Highly soluble in water Low volatility. If applied directly to plants or soil, drift is a concern & mitigation is advisable Persistent in water Highly toxic to birds and honeybees, mason bees, bumble bees & other bee species - <i>Megachile rotundata</i> & <i>Nomia melanderi</i> Toxic to most aquatic species Eye irritant Highly toxic, harmful if swallowed Possible liver toxicant Acute Tox. 1 (100%) Acute Tox. 1 (100%) Acute Tox. 1 (100%) Repr. 1B (50%) STOT RE 1 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H310 (100%): Fatal in contact with skin [Danger Acute toxicity, dermal] H330 (100%): Fatal if inhaled [Danger Acute toxicity, inhalation] H360 (50%): May damage fertility or the unborn child [Danger Reproductive toxicity] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 1 Acute Tox. 1 Acute Tox. 1 STOT RE 1 Aquatic Acute 1 Aquatic Chronic 1 Repr. 1B
60-51-5	XX	Dimethoate	1						1	Potential groundwater contaminant Low aqueous solubility Low volatility. If applied directly to plants or soil, drift is a concern & mitigation is advisable persistent in both soil & water systems Mobile in soils earthworms & mason bees Highly toxic to birds, honeybees & bumble bees & other bee species - <i>Megachile rotundata</i> & <i>Nomia melanderi</i> Toxic to aquatic organisms sediment dwelling organisms Genotoxic - Chromosome aberration (EFSA), Gene mutation (EFSA database) Acetyl cholinesterase inhibitor Respiratory tract, skin & eye irritant Possible liver & prostate toxicant Possible EDC - Endocrine issues - Disruption of thyroid hormones action Severe Marine Pollutant Low volatility Moderately persistent in both soil & water systems Toxic to mammals, earthworms & mason bees Highly toxic to birds, honeybees & bumble bees & other bee species - <i>Megachile rotundata</i> & <i>Nomia melanderi</i> Highly toxic to aquatic organisms Respiratory tract, skin & eye irritant Possible liver & prostate toxicant Possible EDC - Endocrine issues - Disruption of thyroid hormones action Acute Tox. 4 (98.14%) Acute Tox. 4 (98.14%)		H302 (98.14%): Harmful if swallowed [Warning Acute toxicity, oral] H312 (98.14%): Harmful in contact with skin [Warning Acute toxicity, dermal]	Acute Tox. 4 Acute Tox. 4 H302 H312

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110488-70-5	XX	Dimethomorph							Low aqueous solubility Moderately persistent in both soil and water systems Moderately mobile in soils Medium potential for particle bound transport Toxic to earthworms & honey bees Genotoxic - Chromosome aberration (EFA database) Respiratory tract, skin & eye irritant May cause acute lung injury if inhaled Possible liver & prostate toxicant	Low volatility Toxic to aquatic organisms Skin sensitiser	Aquatic Chronic 1 (100%)		H411 (100%): Toxic to aquatic life with long lasting effects [Hazardous to the aquatic environment, long-term hazard]	Aquatic Chronic 2 H411 Repr. 1B H360D
		Dimethomorph + Mancozeb						1	Marine Pollutant solubility Not expected to leach to groundwater persistent in soil & water Highly toxic to fish & aquatic invertebrates & honeybees EDC Genotoxic - Unspecified genotoxicity type Reproductive/developmental toxicant Respiratory tract irritant & skin irritant Possible phytotoxicant Possible carcinogen May cause ovarian hyperthyroidism Possible thyroid toxicant	Quite volatile Not Slightly mobile in soils Toxic to birds, earthworms & honeybees EDC Possible carcinogen	Aquatic Chronic 1 (100%) Skin Sens. 1 (100%) Repr. 2 (90.91%) Aquatic Acute 1 (99.02%) Aquatic Chronic 1 (15.97%)		H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H361 (90.91%): Suspected of damaging fertility or the unborn child [Warning Reproductive toxicity] H400 (99.02%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (15.97%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard] H411 (100%): Toxic to aquatic life with long lasting effects [Hazardous to the aquatic environment, long-term hazard]	Repr. 1B H361 Skin Sens. 1 H317 H351 STOT RE 2 H 37 Aquatic Acute 1 H400 Aquatic Chronic 1 H410 Aquatic Chronic 2 H411
		Dimethyl formamide (N,N - Dimethylformamide)						1	CO-FORMULANT Potential groundwater contaminant Possible drinking water contaminant Low aqueous solubility Low volatility May leach to groundwater Moderately persistent in soil & water Mobile in soils Toxic to birds, earthworms, most aquatic species Highly toxic to algae Toxic to mammals Genotoxic - Chromosome aberration (EFA) EDC - Endocrine issues - Inhibition of androgens action Possible carcinogen Respiratory tract irritant Possible blood, kidney, bladder, spleen and liver toxicant		Acute Tox. 4 (98.68%) Eye Irrit. 2 (98.75%) Acute Tox. 4 (98.47%) Repr. 1B (99.24%)		H312 (98.68%): Harmful in contact with skin [Warning Acute toxicity, dermal] H319 (98.75%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H332 (98.47%): Harmful if inhaled [Warning Acute toxicity, inhalation] H360 (99.24%): May damage fertility or the unborn child [Danger Reproductive toxicity]	Acute Tox. 4 H302 Eye Irrit. 2 H319 Acute Tox. 4 H332 Repr. 1B H360
	Y	Dinoseb						1	Marine Pollutant Rotterdam Convention (Class O) - subject to PIC regulations - ANNEX III CHEMICAL Moderate solubility in water Moderately volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable Non-persistent in soil Mobile in soil High leachability Highly toxic to mammals & birds Highly toxic to fish Toxic to aquatic invertebrates & aquatic crustaceans Occupational Exposure Routes - Readily absorbed through the skin - PPE/PPC required Reproductive / developmental toxicant Skin & eye irritant Possible carcinogen Highly hazardous US EPA - possible human carcinogen		Acute Tox. 2 (70.11%) Acute Tox. 3 (29.89%) Acute Tox. 2 (47.83%) Acute Tox. 3 (52.17%) Skin Irrit. 2 (69.57%) Skin Sens. 1 (69.57%) Eye Dam. 1 (22.28%) Eye Irrit. 2 (77.72%) Acute Tox. 2 (47.83%) Repr. 1B (37.5%) Repr. 1B (62.5%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300+H310+H330 (47.28%): Fatal if swallowed, in contact with skin or if inhaled [Danger Acute toxicity, oral; acute toxicity, dermal; acute toxicity, inhalation] H300 (70.11%): Fatal if swallowed [Danger Acute toxicity, oral] H301+H311 (15.22%): Toxic if swallowed or in contact with skin [Danger Acute toxicity, oral; acute toxicity, dermal] H301 (29.89%): Toxic if swallowed [Danger Acute toxicity, oral] H310 (47.83%): Fatal in contact with skin [Danger Acute toxicity, dermal] H311 (52.17%): Toxic in contact with skin [Danger Acute toxicity, dermal] H315 (69.57%): Causes skin irritation [Warning Skin corrosion/irritation] H317 (69.57%): May cause an allergic skin reaction [Warning Sensitization, Skin] H318 (22.28%): Causes serious eye damage [Danger Serious eye damage/eye irritation] H319 (77.72%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H330 (47.83%): Fatal if inhaled [Danger Acute toxicity, inhalation] H360 (37.5%): May damage fertility or the unborn child [Danger Reproductive toxicity] H360Df (62.5%): May damage the unborn child; Suspected of damaging fertility [Danger Reproductive toxicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 Acute Tox. 3 Eye Irrit. 2 Aquatic Acute 1 Repr. 1B
	XXX		1					1						

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85-00-7	XX	Diquat, including diquat dibromide								Potential groundwater contaminant Highly soluble in water Low risk of leaching to groundwater Volatile Non-mobile in soils High potential for particle bound transport Toxic to mammals Irritant Toxic to honeybees & earthworms Highly toxic to birds Toxic to aquatic organisms Respiratory tract & skin & eye irritant Potential liver, kidney, stomach and intestine toxicant	Met. Corr. 1 (31.2%) Acute Tox. 4 (100%) Skin Irrit. 2 (100%) Skin Sens. 1 (100%) Eye Irrit. 2 (100%) Acute Tox. 2 (100%) STOT SE 3 (100%) STOT RE 1 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H290 (31.2%): May be corrosive to metals [Warning Corrosive to Metals] H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H315 (100%): Causes skin irritation [Warning Skin corrosion/irritation] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H319 (100%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H330 (100%): Fatal if inhaled [Danger Acute toxicity, inhalation] H335 (100%): May cause respiratory irritation [Warning Specific target organ toxicity, single exposure; Respiratory tract irritation] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 *H302 Skin Irrit. 2H315 Eye Irrit. 2H319 Skin Sens. 1H317 Acute Tox. 2 *H330 STOT SE 3H335H335 STOT RE 1H372 Aquatic Acute 1 Aquatic Chronic 1
330-54-1	XX	Diuron								Marine Pollutant (Class Ib) - subject to PIC regulations Strongly phytotoxic Moderately volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable Non-persistent in soils Moderately mobile in soils Toxic to fish, aquatic invertebrates, algae and earthworms Highly toxic to mammals Public exposure routes - Main risk of exposure is thought to be from residues on treated crops Occupational exposure routes - Operator exposure is a concern even with PPE/PPC Genotoxic - Unspecified genotoxicity type Neurotoxicant Skin & eye irritant Skin sensitizer Mutagenic potential; Suspected of causing genetic defects Very toxic by inhalation, skin contact and ingestion Easily absorbed through the skin	Acute Tox. 4 (100%) Carc. 2 (100%) STOT RE 2 (100%) Aquatic Acute 1 (98.17%) Aquatic Chronic 1 (100%)		H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H373 (100%): May cause damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H400 (98.17%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 Carc. 2 STOT RE 2 Aquatic Acute 1 Aquatic Chronic 1
	X	DNOC and salts	1				1			Rotterdam Convention (Class Ib) - subject to PIC regulations - ANNEX III Potential groundwater contaminant Highly soluble in water Very persistent in soil Potential for particle bound transport index Non-mobile in soils Toxic to mammals, honeybees and earthworms Highly toxic to birds Toxic to aquatic organisms Public exposure routes - Slight risk of dietary exposure Respiratory tract , skin & eye irritant Skin sensitizer Potential liver, kidney, stomach and intestine toxicant	Acute Tox. 2 (100%) Acute Tox. 1 (100%) Skin Irrit. 2 (100%) Skin Sens. 1 (100%) Eye Dam. 1 (100%) Acute Tox. 2 (100%) Muta. 2 (100%) Aquatic Acute 1 (97.56%) Aquatic Chronic 1 (100%)		H300+H310+H330 (70.73%): Fatal if swallowed, in contact with skin or if inhaled [Danger Acute toxicity, oral; acute toxicity, dermal; acute toxicity, inhalation] H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H310 (100%): Fatal in contact with skin [Danger Acute toxicity, dermal] H315 (100%): Causes skin irritation [Warning Skin corrosion/irritation] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H318 (100%): Causes serious eye damage [Danger Serious eye damage/eye irritation] H330 (100%): Fatal if inhaled [Danger Acute toxicity, inhalation] H341 (100%): Suspected of causing genetic defects [Warning Germ cell mutagenicity] H400 (97.56%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 Acute Tox. 1 Skin Irrit. 2 Eye Dam. 1 Skin Sens. 1 Acute Tox. 2 Muta. 2 Aquatic Acute 1 Aquatic Chronic 1
115-29-7	XXX	Endosulfan					1	1		Stockholm Convention-POP chemical Subject to PIC regulations - ANNEX III CHEMICAL Marine Pollutant Low solubility in water Low volatility. If applied directly to plants, drift is a concern & mitigation is advisable Moderately volatile Moderately persistent in soil Non-persistent in soil Non-mobile in soil High leachability Bioaccumulates Highly toxic to mammals & other bee species - <i>Trigona spinipes</i> Toxic to birds, earthworms, honey bees & bumble bees Highly toxic to fish & aquatic invertebrates Toxic to aquatic invertebrates, aquatic crustaceans, sediment dwelling organisms & algae Genotoxic-Unspecified genotoxicity type EDC - Endocrine issues - Competitive binding to androgen receptors Reproductive / developmental toxicant Neurotoxicant Mutagenic potential Highly toxic Research shows that endosulfan interferes with the spermatogenic & steroidogenic cycles in rats	Acute Tox. 2 (96.47%) Acute Tox. 4 (96.47%) Acute Tox. 2 (96.47%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300+H330 (32.94%): Fatal if swallowed or if inhaled [Danger Acute toxicity, oral; acute toxicity, inhalation] H300 (96.47%): Fatal if swallowed [Danger Acute toxicity, oral] H312 (96.47%): Harmful in contact with skin [Warning Acute toxicity, dermal] H330 (96.47%): Fatal if inhaled [Danger Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 Acute Tox. 4 Acute Tox. 2 Aquatic Acute 1 Aquatic Chronic 1

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72-20-8	XXX	Endrin								POP - regulated by Stockholm Convention Subject to PIC regulations - CANDIDATE CHEMICAL Severe Marine Pollutant Low solubility in water Moderately volatile Very persistent in soil Non-mobile in soil Low leachability High potential for particle bound transport Bioaccumulative Highly toxic to mammals, birds & honey bees Toxic to earthworms Highly toxic to fish & aquatic invertebrates Toxic to sediment dwelling organisms EDC - Endocrine issues - Competitive binding to androgen receptors Neurotoxicant Possible reproductive / developmental toxicant Highly toxic	Acute Tox. 2 (100%) Acute Tox. 1 (15.79%) Acute Tox. 3 (84.21%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H310 (15.79%): Fatal in contact with skin [Danger Acute toxicity, dermal] H311 (84.21%): Toxic in contact with skin [Danger Acute toxicity, dermal] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 Acute Tox. 3 Aquatic Acute 1 Aquatic Chronic 1
133855-98-8	XX	Epoxiconazole								Low aqueous solubility Persistent in both soil & water systems index Toxic to honeybees, earthworms & most aquatic organisms EDC - Endocrine issues - Inhibition of aromatase activity, decrease of estrogen production Reproductive / developmental toxicant Chromosome aberration (EFSA, Unspecified genotoxicity type Respiratory tract irritant Possible carcinogen thyroid toxicant May cause genetic defects	Carc. 2 (100%) Repr. 1B (69.88%) Repr. 1B (12.36%) Repr. 2 (19.31%) Aquatic Acute 1 (11.58%) Aquatic Chronic 1 (11.58%) Aquatic Chronic 2 (99.23%)		H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H360 (69.88%): May damage fertility or the unborn child [Danger Reproductive toxicity] H360DF (12.36%): May damage the unborn child; Suspected of damaging fertility [Danger Reproductive toxicity] H360FD (18.97%): May damage fertility; May damage the unborn child [Danger Reproductive toxicity] H361 (19.31%): Suspected of damaging fertility or the unborn child [Warning Reproductive toxicity] H400 (11.58%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (11.58%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard] H411 (99.23%): Toxic to aquatic life with long lasting effects [Hazardous to the aquatic environment, long-term hazard]	Carc. 2 H351 Aquatic Chronic 2 H411 Repr. 1B H360DF
		Epoxiconazole & Thiophanate-methyl								POP chemical - Stockholm Convention Chemical Subject to PIC regulations - Rotterdam Convention Annex III Marine Pollutant Low aqueous solubility Volatile Low volatility. If applied directly to plants, drift is a concern & mitigation is advisable Non-mobile in soil Potential for particle bound transport index Tendency to leach to groundwater Bioaccumulation potential honeybees, bumble bees and earthworms Highly toxic to aquatic organisms Highly toxic to mammals & other bee species - Trigona spinipes Genotoxic issues - Competitive binding to androgen receptors Reproductive / developmental toxicant Mutagenic potential Highly toxic	Carc. 2 (100%) Repr. 1B (69.88%) Repr. 1B (12.36%) Repr. 2 (19.31%) Skin Sens. 1 (100%) Acute Tox. 4 (100%) Muta. 2 (100%) Aquatic Acute 1 (11.58%) Aquatic Chronic 1 (11.58%) Aquatic Chronic 2 (99.23%)		H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H332 (100%): Harmful if inhaled [Warning Acute toxicity, inhalation] H341 (100%): Suspected of causing genetic defects [Warning Germ cell mutagenicity] H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H360 (69.88%): May damage fertility or the unborn child [Danger Reproductive toxicity] H360DF (12.36%): May damage the unborn child; Suspected of damaging fertility [Danger Reproductive toxicity] H361 (19.31%): Suspected of damaging fertility or the unborn child [Warning Reproductive toxicity] [Hazardous to the aquatic environment, long-term hazard] H400 (99.5%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (99.5%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard] H411 (99.23%): Toxic to aquatic life with long lasting effects	Carc. 2 Aquatic Chronic 2 Repr. 1B Skin Sens. 1 Acute Tox. 4 Muta. 2 Carc. 2 Aquatic Acute 1 Aquatic Chronic 1
759-94-4	XX	EPTC								Potential groundwater contaminant Highly soluble in water Highly volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable Very persistent in soil systems index Toxic to mammals, birds & aquatic organisms Public exposure routes - Possible risk in some countries via consumption of contaminated drinking water Occupational exposure routes - May be absorbed through inhalation, ingestion, or the skin Carcinogen skin & eye irritant Possible liver and kidney toxicant May cause testicular damage and antispermatic effects High doses may lead to death	Acute Tox. 4 (100%)		H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral]	Acute Tox. 4

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13194-48-4	XX	Ethoprophos	1							Marine Pollutant Highly volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable Bioaccumulates in soils Toxic to fish and aquatic invertebrates Toxic to mammals Genotoxic developmental toxicant Respiratory tract, skin & eye irritant Poinsoning symptoms include fatigue, headache, respiratory problems Blood, lung and liver toxicant	Highly soluble in water Mobile Very toxic to mammals Carcinogen Reproductive / Neurotoxicant Skin sensitiser Poinsoning symptoms include fatigue, headache, respiratory problems Blood, lung and liver toxicant	Acute Tox. 3 (100%) Acute Tox. 1 (100%) Skin Sens. 1 (100%) Acute Tox. 2 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H310 (100%): Fatal in contact with skin [Danger Acute toxicity, dermal] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H330 (100%): Fatal if inhaled [Danger Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 Acute Tox. 1 Skin Sens. 1 Acute Tox. 2 Aquatic Acute 1 Aquatic Chronic 1
		Ethylene dibromide	1							Chemical subject to PIC regulations - ANNEX III Potential groundwater contaminant High solubility in water Highly volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable Very persistent in soil Moderately mobile in soil High leachability Toxic to mammals & birds Toxic to aquatic organisms Public Exposure Routes - Risk of exposure from agricultural sources is negligible Possible risk in some countries via consumption of contaminated drinking water Occupational Exposure Routes-May be absorbed through inhalation, ingestion, or the skin Carcinogen Reproductive / developmental toxicant Skin irritant Respiratory tract irritant Eye irritant Possible EDC & Neurotoxicant Possible liver and kidney toxicant May cause testicular damage and antispermatogenic effects High doses may lead to death	Acute Tox. 3 (100%) Acute Tox. 3 (100%) Skin Irrit. 2 (100%) Eye Irrit. 2 (100%) Acute Tox. 3 (94.12%) STOT SE 3 (100%) Carc. 1B (99.16%) Aquatic Chronic 2 (100%)		H301+H311+H331 (26.05%): Toxic if swallowed, in contact with skin or if inhaled [Danger Acute toxicity, oral; acute toxicity, dermal; acute toxicity, inhalation] H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H311 (100%): Toxic in contact with skin [Danger Acute toxicity, dermal] H315 (100%): Causes skin irritation [Warning Skin corrosion/irritation] H319 (100%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H331 (94.12%): Toxic if inhaled [Danger Acute toxicity, inhalation] H335 (100%): May cause respiratory irritation [Warning Specific target organ toxicity, single exposure; Respiratory tract irritation] H350 (99.16%): May cause cancer [Danger Carcinogenicity] H411 (100%): Toxic to aquatic life with long lasting effects [Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 Acute Tox. 3 Skin Irrit. 2 Eye Irrit. 2 Acute Tox. 3 STOT SE 3 Carc. 1B Aquatic Chronic 2	
75-21-8	XX	Ethylene oxide	1	1			1	1		Persistent-Bioaccumulative-Toxic regulations - ANNEX III Low aqueous solubility Non-mobile in soils in most soils Bioaccumulates earthworms & bumble bees aquatic invertebrates Low volatility Not persistent in water systems Toxic to birds, earthworms & bumble bees Possible liver toxicant	Chemical subject to PIC regulations - ANNEX III Low volatility Not persistent in water systems Toxic to birds, earthworms & bumble bees Possible liver toxicant	Skin Irrit. 2 (40.52%) Skin Sens. 1B (40.52%) Eye Dam. 1 (40.52%) Aquatic Chronic 2 (59.48%) Aquatic Chronic 3 (40.52%)		H315 (40.52%): Causes skin irritation [Warning Skin corrosion/irritation] H317 (40.52%): May cause an allergic skin reaction [Warning Sensitization, Skin] H318 (40.52%): Causes serious eye damage [Danger Serious eye damage/eye irritation] H411 (59.48%): Toxic to aquatic life with long lasting effects [Hazardous to the aquatic environment, long-term hazard] H412 (40.52%): Harmful to aquatic life with long lasting effects [Hazardous to the aquatic environment, long-term hazard]	Press. Gas Flam. Gas 1 Acute Tox. 3 Skin Corr. 1 Eye Dam. 1 Acute Tox. 3 STOT SE 3 Muta. 1B Carc. 1B STOT RE 1 Repr. 1B
		Etoxadole								TRANSFORMATION PRODUCT solubility Persistent in soils Ability to bioaccumulate and aquatic invertebrates to other aquatic organisms, earthworms & honeybees Splene, liver & red blood cell toxicant May cause long term eye damage from repeated exposure Chronic aquatic tox, Transformation product	Volatile Slightly mobile in soils Highly toxic to fish Toxic	Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Aquatic Acute 1 Aquatic Chronic 1
131807-57-3	XX	Famoxadone								Low aqueous solubility Soluble in many organic solvents Not persistent in soil or water systems Mobile in soils Low volatility Bioaccumulative Toxic to earthworms & honey bees Genotoxic - Chromosome aberration Neurotoxicant irritant Splene, liver & red blood cell toxicant May cause long term eye damage from repeated exposure Toxic	Soluble in many organic solvents Not persistent in soil or water systems Low volatility Bioaccumulative Toxic to aquatic organisms Chromosome aberration Neurotoxicant Possible liver and thyroid Splene, liver & red blood cell toxicant May cause long term eye damage from repeated exposure Toxic	STOT RE 2 (100%) Aquatic Acute 1 (85.39%) Aquatic Acute 1 (14.61%) Aquatic Chronic 1 (100%)		H373 (100%): May causes damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H400 (85.39%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H401 (14.61%): Toxic to aquatic life [Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	STOT RE 2 Aquatic Acute 1 Aquatic Chronic 1

		Fenamidone						Potential groundwater contaminant Soluble in water Moderately mobile in soils Highly toxic to mammals, birds & honey bees Genotoxic - Chromosome aberration Neurotoxicant Toxic by inhalation and skin contact Exposure may be fatal due to respiratory failure May overstimulate the nervous system causing nausea, dizziness and confusion Inhibition of erythrocyte ChE activity	Marine Pollutant Low volatility Toxic to earthworms Highly toxic to aquatic organisms Reproductive / developmental toxicant Acetyl cholinesterase inhibitor Eye irritant	Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H400 [100%]: Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 [100%]: Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Aquatic Acute 1 Aquatic Chronic 1
161326-34-7	XX	Fenamiphos						Chemical subject to PIC regulations - Persistent in soil & water systems earthworms & honey bees Genotoxic (chromosome aberration) tract irritant Harmful by inhalation Possible liver toxicant LD50<200mg/kg body weight, Fatal if inhaled - H330 (GHS)	ANNEX III Not mobile in soils Toxic to birds, Highly toxic to aquatic organisms Toxic to mammals Respiratory tract irritant Moderately toxic Respiratory tract irritant Possibly liver toxicant LD50<200mg/kg body weight, Fatal if inhaled - H330 (GHS)	Acute Tox. 2 (100%) Acute Tox. 2 (62.83%) Acute Tox. 3 (37.17%) Eye Irrit. 2 (57.52%) Acute Tox. 2 (62.83%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300+H310+H330 [26.55%]: Fatal if swallowed, in contact with skin or if inhaled [Danger Acute toxicity, oral; acute toxicity, dermal; acute toxicity, inhalation] H300 [100%]: Fatal if swallowed [Danger Acute toxicity, oral] H310 [62.83%]: Fatal in contact with skin [Danger Acute toxicity, dermal] H311 [37.17%]: Toxic in contact with skin [Danger Acute toxicity, dermal] H319 [57.52%]: Causes serious eye irritation [Warning Serious eye damage/eye irritation] H330 [62.83%]: Fatal if inhaled [Danger Acute toxicity, inhalation] H400 [100%]: Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 [100%]: Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 Acute Tox. 2 Eye Irrit. 2 Acute Tox. 2 Aquatic Acute 1 Aquatic Chronic 1
22224-92-6	XX	Fenazaquin	1				1	Low aqueous solubility Persistent in both soil & water systems species, earthworms & honeybees EDC - Endocrine issues - Inhibition of thyroid hormones production Possible reproductive / developmental toxicant Possible liver & kidney toxicant US EPA - possible human carcinogen	Soluble in many organic solvents Non-volatile Non-mobile on soils Toxic to most aquatic species	Acute Tox. 3 (100%) Acute Tox. 4 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H332 [100%]: Harmful if inhaled [Warning Acute toxicity, inhalation] H400 [100%]: Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 [100%]: Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 H301 Acute Tox. 4 H332 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
114369-43-6	XX	Fenbuconazole					1	Severe Marine Pollutant solubility Very persistent in soil & water systems Potential for particle bound transport index volatile Toxic to birds & earthworms toxicant Highly toxic if inhaled	Low aqueous solubility Soluble in several organic solvents Non-mobile in soils Not highly volatile Bioaccumulative Highly toxic to aquatic organisms Reproductive / developmental toxicant Respiratory tract, skin & eye irritant	Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H400 [100%]: Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 [100%]: Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Aquatic Acute 1 Aquatic Chronic 1
		Fenbutatin oxide						Low potential for leaching to groundwater solubility Slightly mobile in soils Toxic to mammals birds, aquatic invertebrates & honeybees, bumble bees & other bee soecies - Andrena erythrionii & Megachile rotundata Acutely toxic to mammals & birds LD50<200mg/kg body weight Toxic to fish, algae & earthworms Exposure Routes - Possible risk when applied inside glasshouses Re-entry after application should be avoided issues - Competitive binding to androgen receptor, inhibition of estrogens action Skin sensitiser damage	Low aqueous solubility Highly soluble in many organic solvents Volatile Bioaccumulates Highly toxic to birds, aquatic invertebrates & honeybees, bumble bees & other bee soecies - Andrena erythrionii & Megachile rotundata Occupational exposure routes - Possible risk when applied inside glasshouses EDC - Endocrine issues - Competitive binding to androgen receptor, inhibition of estrogens action Acetyl cholinesterase inhibitor Skin irritant Risk of serious eye damage	Skin Irrit. 2 (100%) Eye Irrit. 2 (100%) Acute Tox. 2 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H315 (100%): Causes skin irritation [Warning Skin corrosion/irritation] H319 (100%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H330 [100%]: Fatal if inhaled [Danger Acute toxicity, inhalation] H400 [100%]: Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 [100%]: Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Skin Irrit. 2 Eye Irrit. 2 Acute Tox. 2 Aquatic Acute 1 Aquatic Chronic 1
13356-08-6	XX	Fenitrothion					1	Severe Marine Pollutant Moderately volatile Potential for particle bound transport index Persistent in both soil systems & water bodies Highly toxic to bees, fish & aquatic invertebrates Toxic to birds, earthworms & algae mammals & birds LD50<200mg/kg body weight humans via the oral route Highly toxic	Low aqueous solubility Non-mobile in soils Bioaccumulates Toxic to birds, earthworms & algae Toxic to humans via the oral route Respiratory tract irritant	Acute Tox. 4 (60%) Acute Tox. 4 (40%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (64.14%)		H302 (60%): Harmful if swallowed [Warning Acute toxicity, oral] H312 (40%): Harmful in contact with skin [Warning Acute toxicity, dermal] H400 [100%]: Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 [64.14%]: Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 Aquatic Acute 1 Aquatic Chronic 1
122-14-5	XX						1						

UnPoison South Africa's Highly Hazardous Pesticide Database vs EU's According to JMPM Criteria (FAO/WHO), GHS and ECHA Classification Systems - February 2025

39515-41-8	XX	Fenpropathrin								1	Severe Marine Pollutant subject to PIC regulations - RECOMMENDED FOR LISTING Low water solubility organic solvents Persistent in soil & water systems Bioaccumulates earthworms honeybees Highly toxic to aquatic invertebrates, aquatic crustaceans & sediment dwelling organisms Routes - Ingestion main poisoning route Genotoxic - Unspecified genotoxicity type (EFSA) Acetyl cholinesterase inhibitor Chemical Highly soluble in Low volatility Slightly mobile in soils Toxic to mammals & earthworms Toxic to fish & algae Occupational Exposure	Respiratory tract, skin & eye irritant		H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H312 (100%): Harmful in contact with skin [Warning Acute toxicity, dermal] H330 (100%): Fatal if inhaled [Danger Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 *H301 Acute Tox. 4 *H312 Acute Tox. 2 *H330 Aquatic Acute 1H400 Aquatic Chronic 1 H410
55-38-9	XX	Fenthion								1	Obsolete organotin fungicide Pollutant aqueous solubility Persistent in water persistent in soils Low leachability Toxic to mammals, birds & earthworms toxic to mammals via the oral route Occupational Exposure Routes - Occupational exposure may occur through inhalation and dermal contact Respiratory tract irritant Reproductive / developmental toxicant, skin irritant & eye irritant CLP data - suspected carcinogen probable human carcinogen system, cardiovascular, and respiratory systems Severe Marine Low Low volatility Moderately Slightly mobile in soils Bioaccumulative Highly toxic to Moderately Acute Tox. 3 (12.73%) Acute Tox. 4 (87.27%) Acute Tox. 2 (10.91%) Acute Tox. 4 (85.45%) Acute Tox. 3 (98.18%) Muta. 2 (100%) STOT RE 1 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H301 (12.73%): Toxic if swallowed [Danger Acute toxicity, oral] H302 (87.27%): Harmful if swallowed [Warning Acute toxicity, oral] H310 (10.91%): Fatal in contact with skin [Danger Acute toxicity, dermal] H312 (85.45%): Harmful in contact with skin [Warning Acute toxicity, dermal] H331 (98.18%): Toxic if inhaled [Danger Acute toxicity, inhalation] H341 (100%): Suspected of causing genetic defects [Warning Germ cell mutagenicity] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 Acute Tox. 4 Acute Tox. 3 Muta. 2 STOT RE 1 Aquatic Acute 1 Aquatic Chronic 1	
76-87-9	XX	Fentin hydroxide								1	Low aqueous solubility Mobile in soil systems Highly toxic to humans and birds, honey bees, other bee species - Megachile rotundata & Melipona scutellaris earthworms Neurotoxicant Eye irritant Thyroid, kidney and liver toxicant Highly toxic Bioaccumulates US EPA - possible human carcinogen genetic defects Low volatility Quite persistent in soil Slightly mobile May bio-accumulate Highly toxic to mammals Toxic to Toxic to aquatic organisms Skin irritant Possible carcinogen, EDC, Reproductive / developmental toxicant Suspected of causing Acute Tox. 3 (100%) Acute Tox. 3 (100%) Skin Irrit. 2 (100%) Eye Dam. 1 (100%) Acute Tox. 2 (100%) STOT SE 3 (100%) Carc. 2 (100%) Repr. 2 (63.64%) Repr. 2 (36.36%) STOT SE 2 (98.7%) STOT RE 1 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H301+H311 (85.71%): Toxic if swallowed or in contact with skin [Danger Acute toxicity, oral; acute toxicity, dermal] H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H311 (100%): Toxic in contact with skin [Danger Acute toxicity, dermal] H315 (100%): Causes skin irritation [Warning Skin corrosion/irritation] H318 (100%): Causes serious eye damage [Danger Serious eye damage/eye irritation] H330 (100%): Fatal if inhaled [Danger Acute toxicity, inhalation] H335 (100%): May cause respiratory irritation [Warning Specific target organ toxicity, single exposure; Respiratory tract irritation] H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H361 (63.64%): Suspected of damaging fertility or the unborn child [Warning Reproductive toxicity] H361d (36.36%): Suspected of damaging the unborn child [Warning Reproductive toxicity] H371 (98.7%): May cause damage to organs [Warning Specific target organ toxicity, single exposure] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 Acute Tox. 3 Skin Irrit. 2 Eye Dam. 1 Acute Tox. 2 STOT SE 3 Carc. 2 STOT RE 1 Aquatic Acute 1 Aquatic Chronic 1 Repr. 2	

UnPoison South Africa's Highly Hazardous Pesticide Database vs EU's According to JMPM Criteria (FAO/WHO), GHS and ECHA Classification Systems - February 2025

120068-37-3	XX	Fipronil							Coumarin rodenticide -second-generation indirect anticoagulant Low aqueous solubility Low volatility Persistent in both soil & water systems May leach to groundwater Slightly mobile in soils High potential for particle bound transport Bioaccumulates Toxic to birds, aquatic invertebrates & algae Highly toxic to fish & mammals Acutely toxic to mammals & birds LD50<200mg/kg body weight Toxic to bees High mammalian oral toxicity Occupational Exposure Routes - May be absorbed through the skin Occupational exposure should be low or non-existent since flocoumafen is no longer used in most countries Possible reproduction / developmental toxicant Highly toxic May cause massive hemorrhage resulting in shock	Acute Tox. 3 (100%) Acute Tox. 3 (100%) Acute Tox. 2 (45.66%) Acute Tox. 3 (92.49%) STOT RE 1 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H311 (100%): Toxic in contact with skin [Danger Acute toxicity, dermal] H330 (45.66%): Fatal if inhaled [Danger Acute toxicity, inhalation] H331 (92.49%): Toxic if inhaled [Danger Acute toxicity, inhalation] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 Acute Tox. 3 STOT RE 1 Aquatic Acute 1 Aquatic Chronic 1
90035-08-8	XX	Flocoumafen	1						Low aqueous solubility Miscible in many organic solvents Non-volatile Low volatility. If applied directly to plants, drift is a concern & mitigation is advisable Volatilisation is potentially a concern for short and long range air transport Potential for particle bound transport index Not persistent in soil or water Unlikely to leach to groundwater Slightly Mobile in soil Bioaccumulation concerns Toxic to earthworms & aquatic organisms Reproductive / developmental toxicant Skin sensitiser Respiratory tract & skin irritant Possible Harmful if inhaled or ingested Kidney, spleen and liver toxicant May cause eye cataracts acutely toxic to mammals & birds LD50<200mg/kg body weight, Fatal if inhaled - H330 (GHS)	Acute Tox. 2 (100%) Acute Tox. 1 (100%) Acute Tox. 2 (100%) STOT RE 1 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H310 (100%): Fatal in contact with skin [Danger Acute toxicity, dermal] H330 (100%): Fatal if inhaled [Danger Acute toxicity, inhalation] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 Acute Tox. 1 Acute Tox. 2 STOT RE 1 Aquatic Acute 1 Aquatic Chronic 1
		Fluazifop-butyl				1		1	Low aqueous solubility Non-volatile Moderately persistent in soils Non-mobile in soils High potential for particle bound transport Highly toxic to most aquatic organisms Toxic to birds, honeybees & earthworms Respiratory tract irritant EDC May cause anemia Possible blood, liver, kidney and spleen toxicant	Repr. 1B (58.57%) Repr. 1B (40%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H360 (58.57%): May damage fertility or the unborn child [Danger Reproductive toxicity] H360D (40%): May damage the unborn child [Danger Reproductive toxicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Aquatic Acute 1 Aquatic Chronic 1 Repr. 1B
		Flufenoxuron				1		1	Low aqueous solubility Relatively volatile Low volatility. If applied directly to plants, drift is a concern & mitigation is advisable Not persistent in soils or water Slightly mobile in soils High risk of bioconcentration Highly toxic to algae and aquatic plants Toxic to earthworms, honeybees, fish and aquatic invertebrates Low mammalian toxicity Occupational exposure route - Occupational exposure may occur through dermal and inhalation routes Possible Reproductive/ developmental toxicant Possible Respiratory tract irritant Eye irritant Some evidence chronic exposure can cause anaemia	Acute Tox. 4 (31.43%) Lact. (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H332 (31.43%): Harmful if inhaled [Warning Acute toxicity, inhalation] H362 (100%): May cause harm to breast-fed children [Reproductive toxicity, effects on or via lactation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Lact. H362 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
		Flumioxazin						1	Low solubility in water Low volatility. If applied directly to plants, drift is a concern & mitigation is advisable Non-Volatile Slightly mobile in soil Potential for particle bound transport Bioaccumulation concerns Toxic to earthworms & honey bees Toxic to aquatic organisms Reproductive / developmental toxicant Skin sensitiser Possible liver and testes toxicant	Repr. 1B (39.15%) Repr. 2 (60.85%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H360 (39.15%): May damage fertility or the unborn child [Danger Reproductive toxicity] H361d (60.85%): Suspected of damaging the unborn child [Warning Reproductive toxicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Aquatic Acute 1 Aquatic Chronic 1 Repr. 2

UnPoison South Africa's Highly Hazardous Pesticide Database vs EU's According to JMPM Criteria (FAO/WHO), GHS and ECHA Classification Systems - February 2025

		Fluorochloridone								1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															</
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UnPoison South Africa's Highly Hazardous Pesticide Database vs EU's According to JMPM Criteria (FAO/WHO), GHS and ECHA Classification Systems - February 2025

50-00-0	XXY	Formaldehyde								CO-FORMULANT EC 2232/96 Annex 1 registered as food flavouring agent Highly soluble in water Highly volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable Moderately volatile Non-persistent in soils Low leachability Mobile in soils Toxic to mammals, birds & honey bees Toxic to aquatic organisms Occupational Exposure Routes - Exposure can occur via inhalation, ingestion, skin contact, and absorption through the skin Carcinogen Neurotoxicant Respiratory tract, skin & eye irritant Skin sensitiser Highly toxic	Acute Tox. 3 (83.48%) Acute Tox. 3 (83.48%) Skin Corr. 1B (83.48%) Skin Sens. 1 (90.66%) Eye Dam. 1 (47.65%) Acute Tox. 3 (78.85%) Carc. 2 (65.2%)		H301 (83.48%): Toxic if swallowed [Danger Acute toxicity, oral] H311 (83.48%): Toxic in contact with skin [Danger Acute toxicity, dermal] H314 (83.48%): Causes severe skin burns and eye damage [Danger Skin corrosion/irritation] H317 (90.66%): May cause an allergic skin reaction [Warning Sensitization, Skin] H318 (47.65%): Causes serious eye damage [Danger Serious eye damage/eye irritation] H331 (78.85%): Toxic if inhaled [Danger Acute toxicity, inhalation] H351 (65.2%): Suspected of causing cancer [Warning Carcinogenicity]	Acute Tox. 3 Eye Irrit. 2 STOT SE 3 Skin Corr. 1B Skin Irrit. 2 Carc. 1B
		Formetanate	1						1	Marine Pollutant High solubility in water Low volatility Non-persistent in soil & water Moderately mobile in soil Low leachability Highly toxic to mammals, birds & honey bees Toxic to earthworms & bumble bees Highly toxic to aquatic invertebrates Toxic to fish, sediment dwelling organisms & algae Occupational Exposure Routes-Possible exposure via dermal or inhalation route Acetyl cholinesterase inhibitor Eye irritant Highly toxic	Acute toxicity - category 2 Acute toxicity - category 2 Skin sensitisation - category 1 Hazardous to the aquatic environment (acute) - category 1 Hazardous to the aquatic environment (chronic) - category		H300: Fatal if swallowed [Danger Acute toxicity, oral] H317: May cause an allergic skin reaction [Warning Sensitization, Skin] H330: Fatal if inhaled [Danger Acute toxicity, inhalation] H410: Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 Skin Sens. 1 Acute Tox. 2 Aquatic Acute 1 Aquatic Chronic 1
1/1/1998	XX	Furfural		1					1	Highly soluble in water Volatile Low risk of leaching to groundwater Moderately toxic to mammals Reproductive / development effects Neurotoxicant Harmful if ingested, inhaled or absorbed through the skin Possible kidney, bladder, blood and lungs toxicant	Acute Tox. 3 (100%) Acute Tox. 4 (99.95%) Skin Irrit. 2 (94.99%) Eye Irrit. 2A (100%) Acute Tox. 2 (12.26%) Acute Tox. 3 (87.79%) STOT SE 3 (100%) Carc. 2 (100%)		H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H312 (99.95%): Harmful in contact with skin [Warning Acute toxicity, dermal] H315 (94.99%): Causes skin irritation [Warning Skin corrosion/irritation] H319 (100%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H330 (12.26%): Fatal if inhaled [Danger Acute toxicity, inhalation] H331 (87.79%): Toxic if inhaled [Danger Acute toxicity, inhalation] H335 (100%): May cause respiratory irritation [Warning Specific target organ toxicity, single exposure; Respiratory tract irritation] H351 (100%): Suspected of causing cancer [Warning Carcinogenicity]	Carc. 1B

UnPoison South Africa's Highly Hazardous Pesticide Database vs EU's According to JMPM Criteria (FAO/WHO), GHS and ECHA Classification Systems - February 2025

	Glufofenate				1			1	Strong acid Toxic to mammals Reproductive / developmental effects Neurotoxicant Possible Skin Irritant & Eye irritant	Acute Tox. 4 (100%) Acute Tox. 4 (97.8%) Acute Tox. 4 (97.8%) Repr. 1B (35.1%) Repr. 1B (63.2%) STOT SE 1 (62.7%) STOT RE 2 (97.8%) Aquatic Acute 1 (62.7%) Aquatic Chronic 1 (62.7%)		H302+H312+H332 (63.8%): Harmful if swallowed, in contact with skin or if inhaled [Warning Acute toxicity, oral; acute toxicity, dermal; acute toxicity, inhalation] H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H312 (97.8%): Harmful in contact with skin [Warning Acute toxicity, dermal] H332 (97.8%): Harmful if inhaled [Warning Acute toxicity, inhalation] H360 (35.1%): May damage fertility or the unborn child [Danger Reproductive toxicity] H360Fd (63.2%): May damage fertility; Suspected of damaging the unborn child [Danger Reproductive toxicity] H370 (62.7%): Causes damage to organs [Danger Specific target organ toxicity, single exposure] H373 (97.8%): May causes damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H400 (62.7%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (62.7%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 H302 Acute Tox. 4 H312 Acute Tox. 4 H332 STOT RE 2 H373 Repr. 1B H360Fd
	Glufofenate-ammonium				1			1	Risk of herbicide resistance developing - anti-resistance management required Highly soluble in water Persistent in aquatic systems Relatively volatile Slightly mobile in soils Low leachability Toxic to birds, most aquatic organisms, earthworms and honeybees Chrysoperla carnea Larva, Beneficial insects (Parasitic wasps) - Aphidius rhopalosiphii Adult & Beneficial insects (Predatory mites) - Typhlodromus pyri Adult Toxic to aquatic crustaceans & aquatic plants Public Exposure Routes-Slight risk to bystanders in orchards Occupational Exposure Routes-Risk of exposure acceptable under label recommendations for use for personal protection clothing and equipment Toxic to humans Reproductive / developmental toxicant Neurotoxicant Possible skin& eye irritant Harmful if ingested, inhaled or absorbed through the skin Possible kidney, bladder, blood and lungs toxicant	Acute Tox. 4 (100%) Acute Tox. 4 (97.81%) Acute Tox. 4 (97.81%) Repr. 1B (34.97%) Repr. 1B (63.39%) STOT SE 1 (63.39%) STOT RE 2 (100%) Aquatic Acute 1 (63.39%) Aquatic Chronic 1 (63.39%)		H302+H312+H332 (63.39%): Harmful if swallowed, in contact with skin or if inhaled [Warning Acute toxicity, oral; acute toxicity, dermal; acute toxicity, inhalation] H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H312 (97.81%): Harmful in contact with skin [Warning Acute toxicity, dermal] H332 (97.81%): Harmful if inhaled [Warning Acute toxicity, inhalation] H360 (34.97%): May damage fertility or the unborn child [Danger Reproductive toxicity] H360Fd (63.39%): May damage fertility; Suspected of damaging the unborn child [Danger Reproductive toxicity] H370 (63.39%): Causes damage to organs [Danger Specific target organ toxicity, single exposure] H373 (100%): May causes damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H400 (63.39%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (63.39%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 Acute Tox. 4 Acute Tox. 4 STOT RE 2 Repr. 1B
	Glyphosate				1			1	Low solubility in water Relatively volatile High persistent in soils Reproductive / developmental toxicant Respiratory tract irritant neurotoxicant Possible Harmful if swallowed May cause haematological changes	Acute Tox. 4 (100%) Acute Chronic 2 (100%)		H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H411 (100%): Toxic to aquatic life with long lasting effects [Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 Aquatic Chronic 2 Eye Dam. 1
	Halosulfuron-methyl				1			1	Severe Marine Pollutant Low aqueous solubility Volatile Low potential to leach to groundwater Persistent in soil systems Non-Mobile in soils May bioaccumulate Toxic to mammals and birds Highly toxic to honeybees, other bee species Trigona spinipes and aquatic organisms EDC Endocrine issues Binding to cellular estrogen and androgen receptors Neurotoxicant Reproductive / developmental toxicant Eye irritant Liver toxicant	Reproductive toxicity - Category 1B Hazardous to the aquatic environment (Acute) - Category 1 Hazardous to the aquatic environment (Long-term) - Category 1		H360: May damage fertility or the unborn child [Danger Reproductive toxicity] H400: Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410: Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Aquatic Acute 1 Aquatic Chronic 1 Repr. 1B

UnPoison South Africa's Highly Hazardous Pesticide Database vs EU's According to JMPM Criteria (FAO/WHO), GHS and ECHA Classification Systems - February 2025

		Heptachlor								Rotterdam Convention (Class 1a) - subject to PIC regulations - ANNEX III Listed under Stockholm convention - POP Volatile applied directly to plants & soil, drift is a concern & mitigation is advisable Very persistent in soils High potential to bioaccumulate Highly toxic to fish Toxic to birds, aquatic invertebrates, algae & earthworms Carcinogen EDC - Endocrine issues - Severe disruption of thyroid hormone production Skin irritant reproductive/ developmental toxicant Extremely hazardous	Acute Tox. 2 (93.62%) Acute Tox. 2 (93.62%) Carc. 2 (100%) STOT RE 2 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300 (93.62%): Fatal if swallowed [Danger Acute toxicity, oral] H310 (93.62%): Fatal in contact with skin [Danger Acute toxicity, dermal] H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H373 (100%): May causes damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 Acute Tox. 3 Carc. 2 STOT RE 2 Aquatic Acute 1 Aquatic Chronic 1
76-44-8	XXX	Hexachlorobenzene					1	1	1	Rotterdam Convention (Class 1a) - subject to PIC regulations - ANNEX III Listed under Stockholm convention - POP Moderate solubility Low volatility. If applied directly to plants, drift is a concern & mitigation is advisable Non-volatile non-mobile Persistent in soil & Toxic to mammals, birds, honeybees & bumble bees aquatic organisms Carcinogen EDC - Endocrine issues - Severe disruption of thyroid hormone production Skin irritant Possible reproductive/ developmental toxicant Extremely hazardous	Carc. 1B (100%) STOT RE 1 (100%) Aquatic Acute 1 (98.35%) Aquatic Chronic 1 (100%)		H350 (100%): May cause cancer [Danger Carcinogenicity] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H400 (98.35%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Carc. 1B STOT RE 1 Aquatic Acute 1 Aquatic Chronic 1
608-73-1	XXX	Hydramethylinon		1	1			1	1	Low solubility in water Moderately volatile in soils & water Low leachability for particle bound transport Bioaccumulates Non-persistent Non-mobile in soils Medium potential Toxic to mammals, birds & honeybees organisms Highly toxic aquatic invertebrates Occupational exposure may occur through inhalation and dermal contact Reproductive / developmental toxicant Respiratory tract, eye & skin irritant Possible carcinogen - US EPA	Acute Tox. 4 (100%) Eye Irrit. 2 (100%) Acute Tox. 4 (56.5%) Repr. 1B (74.7%) STOT RE 1 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H319 (100%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H332 (56.5%): Harmful if inhaled [Warning Acute toxicity, inhalation] H360 (74.7%): May damage fertility or the unborn child [Danger Reproductive toxicity] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 Eye Irrit. 2 STOT RE 1 Aquatic Acute 1 Aquatic Chronic 1
67485-29-4	XX	Imazalil				1				Neonicotinoid insecticide Non-volatile Moderately mobile particle bound transport bioaccumulating Highly soluble Persistent in soil Potential risk for Low risk of Highly toxic to birds, honeybees, bumble bees, mason bees & other bee species - Megachile rotundata earthworms & other bees species -Scaptotrigona postica Toxic to aquatic organisms Genotoxic - Unspecified genotoxicity type Reproductive / developmental toxicant Possibly neurotoxicant, skin irritant & eye irritant Moderately toxic Potential liver, kidney, thyroid, heart and spleen toxicant	Acute Tox. 3 (71.43%) Acute Tox. 4 (28.57%) Eye Dam. 1 (100%) Acute Tox. 4 (100%) Carc. 2 (68%) STOT RE 2 (20%) Aquatic Acute 1 (53.14%) Aquatic Chronic 1 (100%)		H301 (71.43%): Toxic if swallowed [Danger Acute toxicity, oral] H302 (28.57%): Harmful if swallowed [Warning Acute toxicity, oral] H318 (100%): Causes serious eye damage [Danger Serious eye damage/eye irritation] H332 (100%): Harmful if inhaled [Warning Acute toxicity, inhalation] H351 (68%): Suspected of causing cancer [Warning Carcinogenicity] H373 (20%): May causes damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H400 (53.14%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 Eye Dam. 1 Acute Tox. 4 Carc. 2 Aquatic Chronic 1
138261-41-3	XX	Imidacloprid							1	Highly soluble Persistent in the soil mobile in soil Potential for particle bound transport index Highly toxic to birds & honeybees, masonbees, bumble bees, other bee species - Megachile rotundata & Scaptotrigona postica Toxic to mammals & earthworms Acutely toxic to mammals & birds LD50<200mg/kg body weight Toxic to aquatic organisms Genotoxic Aquatic toxicity LC/EC5-<50µg/l Reproductive / developmental toxicant	Acute Tox. 3 (34.26%) Acute Tox. 4 (65.74%) Aquatic Acute 1 (95.82%) Aquatic Chronic 1 (100%)		H301 (34.26%): Toxic if swallowed [Danger Acute toxicity, oral] H302 (65.74%): Harmful if swallowed [Warning Acute toxicity, oral] H400 (95.82%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 Aquatic Acute 1 Aquatic Chronic 1

UnPoison South Africa's Highly Hazardous Pesticide Database vs EU's According to JMPM Criteria (FAO/WHO), GHS and ECHA Classification Systems - February 2025

173584-44-6	XX	Indoxacarb							1	Marine Pollutant High solubility in water Non-persistent in soil Highly toxic to birds Organisms Public exposure route - Consumers may be exposed via ingestion of contaminated food Occupational exposure route - May be absorbed through the skin EDC Possible reproductive / developmental toxicant Possible thyroid toxicant May cause gastroenteritis, fever and hyperthermia Non-volatile Moderately mobile in soil Toxic to mammals & earthworms Toxic to aquatic organisms Eye irritant	Acute Tox. 3 (100%) Skin Sens. 1B (100%) Acute Tox. 4 (100%) STOT RE 1 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H332 (100%): Harmful if inhaled [Warning Acute toxicity, inhalation] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 Skin Sens. 1B Acute Tox. 4 STOT RE 1 Aquatic Acute 1 Aquatic Chronic 1
1689-83-4	XX	Ioxynil							1	Low solubility in water Persistent in soil Persistent in water Potential for particle bound transport index Genotoxic - Chromosome aberration (EFSA) Possible stomach toxicant May cause eye damage Non-volatile Moderately mobile in soil Slightly mobile in soil Bioaccumulation potential Toxic to mammals, birds & earthworms Genotoxic - Possible reproductive /	Acute Tox. 3 (100%) Acute Tox. 4 (100%) Eye Irrit. 2 (100%) Acute Tox. 3 (100%) Repr. 2 (62.16%) Repr. 2 (37.84%) STOT RE 2 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H301+H331 (37.84%): Toxic if swallowed or if inhaled [Danger Acute toxicity, oral; acute toxicity, inhalation] H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H312 (100%): Harmful in contact with skin [Warning Acute toxicity, dermal] H319 (100%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H331 (100%): Toxic if inhaled [Danger Acute toxicity, inhalation] H361 (62.16%): Suspected of damaging fertility or the unborn child [Warning Reproductive toxicity] H361d (37.84%): Suspected of damaging the unborn child [Warning Reproductive toxicity] H373 (100%): May causes damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 Acute Tox. 4 Eye Irrit. 2 Acute Tox. 3 STOT RE 2 Aquatic Acute 1 Aquatic Chronic 1 Repr. 2
125225-28-7	XX	Ipronazole							1	Low water solubility Not persistent in soils and water Organisms EDC - Endocrine issues - Increase weakly aromatase activity Reproductive /developmental toxicant Irritant Possible liver, adrenals, testes, postrate & spleen toxicant Hepatotoxic in mice US EPA - likely to be human carcinogen; CLP data - suspected carcinogen Non-volatile Slightly mobile in soils Toxic to earthworms & aquatic organisms Genotoxic - Unsuspected genotoxicity type Respiratory tract May cause pulmonary problems	Acute Tox. 4 (83.55%) Acute Tox. 4 (32.47%) Repr. 1B (52.38%) Repr. 2 (14.72%) STOT RE 2 (60.17%) Aquatic Acute 1 (23.38%) Aquatic Chronic 1 (76.19%)		H302 (83.55%): Harmful if swallowed [Warning Acute toxicity, oral] H332 (32.47%): Harmful if inhaled [Warning Acute toxicity, inhalation] H360D (52.38%): May damage the unborn child [Danger Reproductive toxicity] H361 (14.72%): Suspected of damaging fertility or the unborn child [Warning Reproductive toxicity] H373 (60.17%): May causes damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H400 (23.38%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (76.19%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 STOT RE 2 Aquatic Chronic 1 Repr. 1B
36734-19-7	XX	Iprodione							1	Marine pollutant Rotterdam Convention (Class 1a) - subject to PIC regulations - RECOMMENDED FOR LISTING Low aqueous solubility Semi-volatile Risk may leach to groundwater Non-persistent in soil Mobile in soils Persists in water for a long time Toxic to birds, most aquatic organisms and earthworms Moderately toxic to mammals Possible carcinogen	Carcinogenicity - category 2 Hazardous to the aquatic environment (acute) - category 1 Hazardous to the aquatic environment (chronic) - category 1		H351: Suspected of causing cancer [Warning Carcinogenicity] H410: Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Carc. 2 Aquatic Acute 1 Aquatic Chronic 1
		Iprovalicarb							1	Phytotoxic Non-persistent in soils & water Toxic to aquatic organisms carcinogen developmental toxicant Possible neurotoxicant Low volatility Moderately mobile in soils Toxic to earthworms Possible reproductive /	Low solubility in water Low volatility Moderately mobile in soils Toxic to earthworms Possible reproductive /		H351: Suspected of causing cancer [Warning Carcinogenicity]	Carc. 2

	Isoxafutole									POP (recommended for Annex I of POPs Treaty) Air pollutant Solubility Non-volatile Low aqueous Very persistent in soil & water & air Slightly mobile Potential for particle bound transport High potential to bioaccumulate Toxic to birds & earthworms Highly toxic to most aquatic organisms Highly toxic to mammals via the oral route Public exposure routes - Food residues were a problem prior to the substance being banned May be absorbed via ingestion Occupational exposure routes - A large number of cases of occupational poisoning have been reported Possible carcinogen Estrogen and androgen receptors EDC - Endocrine issues - Binding to Respiratory tract, Skin & eye irritant Moderately toxic Possible liver & kidney toxicant May cause severe tremor	Repr. 2 (38.83%) Repr. 2 (61.7%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H361 (38.83%): Suspected of damaging fertility or the unborn child [Warning Reproductive toxicity] H361d (61.7%): Suspected of damaging the unborn child [Warning Reproductive toxicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Aquatic Acute 1 Repr. 2
X	Kepone								1	Listed under the Stockholm Convention - POP Low solubility in water Non-volatile Non-persistent in soil Moderately mobile in soil persistent in water Bioconcentration threshold for concern Toxic to earthworms & aquatic organisms Respiratory tract, skin & eye irritant Possible carcinogen May damage esophagus or gastrointestinal tract Possible liver toxicant	Acute Tox. 3 (100%) Carc. 2 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H311 (100%): Toxic in contact with skin [Danger Acute toxicity, dermal] H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 H301 Acute Tox. 3 H311 Carc. 2 H351 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
	Kresoxim-methyl								1	Low aqueous solubility Relatively volatile Nonmobile Persistent in soils Potential for particle bound transport Bioaccumulative Highly toxic to fish, aquatic invertebrates and honeybees, mason bees & bumble bees Toxic to earthworms Highly toxic to mammals Genotoxic - DNA damage/repair (EFSA) Respiratory tract & eye irritant Skin sensitizer Possible reproductive / developmental toxicant Harmful if swallowed, inhaled or in contact with skin Possible neurotoxicant Possible immune system and thyroid toxicant in susceptible individuals	Carc. 2 (100%) Aquatic Acute 1 (64.08%) Aquatic Chronic 1 (100%)		H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H400 (64.08%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Carc. 2 Aquatic Acute 1 Aquatic Chronic 1
	Lambda-cyhalothrin								1	Low water solubility Low volatility. If applied directly to plants or soil, drift is a concern & mitigation is advisable Persistent in soils Very persistent in soils Non-mobile in soils High potential for biocummulation Invertebrates Highly toxic to mammals, fish & aquatic invertebrates Toxic to birds Occupational Exposure Routes - May occur through dermal contact and inhalation of dust and sprays Reproductive / developmental toxicant Acetyl cholinesterase inhibitor Neurotoxicant Eye irritant Highly toxic	Acute Tox. 3 (100%) Acute Tox. 3 (24.62%) Acute Tox. 4 (75%) Eye Irrit. 2 (13.85%) Acute Tox. 2 (100%) STOT SE 1 (13.46%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H301+H311 (11.15%): Toxic if swallowed or in contact with skin [Danger Acute toxicity, oral; acute toxicity, dermal] H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H311 (24.62%): Toxic in contact with skin [Danger Acute toxicity, dermal] H312 (75%): Harmful in contact with skin [Warning Acute toxicity, dermal] H319 (13.85%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H330 (100%): Fatal if inhaled [Danger Acute toxicity, inhalation] H370 (13.46%): Causes damage to organs [Danger Specific target organ toxicity, single exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 Acute Tox. 4 Acute Tox. 2 Aquatic Acute 1 Aquatic Chronic 1
X	Leptophos								1	CO-FORMULANT Mutagenic/ genotoxic Irritant Respiratory tract May cause eye damage Carcinogenic	Acute Tox. 3 (92%) Acute Tox. 4 (92%) STOT SE 1 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H301 (92%): Toxic if swallowed [Danger Acute toxicity, oral] H312 (92%): Harmful in contact with skin [Warning Acute toxicity, dermal] H370 (100%): Causes damage to organs [Danger Specific target organ toxicity, single exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 Acute Tox. 4 STOT SE 1 Aquatic Acute 1 Aquatic Chronic 1

UnPoison South Africa's Highly Hazardous Pesticide Database vs EU's According to JMPM Criteria (FAO/WHO), GHS and ECHA Classification Systems - February 2025

		Light aromatic solvent																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															
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UnPoison South Africa <https://unpoison.org>

UnPoison South Africa's Highly Hazardous Pesticide Database vs EU's According to JMPM Criteria (FAO/WHO), GHS and ECHA Classification Systems - February 2025

	MCPA (potassium salt)								Potential groundwater contaminant Highly soluble in water Low volatility. If applied directly to plants, drift is a concern & mitigation is advisable Non-persistent in soil & water Mobile in soil Bioaccumulates Toxic to mammals, birds & earthworms Toxic to aquatic plants Public Exposure Routes - May be absorbed through the skin Occupational Exposure Routes - Occupational exposure may occur through inhalation and dermal contact Genotoxic - Unspecified genotoxicity type Eye irritant Possible reproductive/developmental toxicant, respiratory tract irritant & skin irritant Highly toxic May cause hypotension Possible liver toxicant	Acute Tox. 4 (100%)	Acute Tox. 4 (100%)	Acute Tox. 4 (100%)	Aquatic Acute 1 (49.42%)	Aquatic Chronic 1 (49.42%)		H302+H312+H332 (16.28%): Harmful if swallowed, in contact with skin or if inhaled [Warning Acute toxicity, oral; acute toxicity, dermal; acute toxicity, inhalation] H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H312 (100%): Harmful in contact with skin [Warning Acute toxicity, dermal] H332 (100%): Harmful if inhaled [Warning Acute toxicity, inhalation] H400 (49.42%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (49.42%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4												
	MCPA (sodium salt)								Low aqueous solubility Low volatility Unlikely to leach to groundwater Not persistent in soil Moderately persistent in water Non-mobile in soil Potential for particle bound transport Bioaccumulation a concern Toxic to mammals, earthworms & honeybees Highly toxic to fish & aquatic invertebrates Toxic to algae Reproductive/developmental toxicant Skin & eye irritant Skin sensitiser Possible liver and thyroid toxicant	Acute Tox. 4 (100%)			H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral]	Acute Tox. 4															
	Meptyldinocap								Low aqueous solubility Volatile Moderately Not persistent in soil Non-mobile in soils Low leachability Medium potential for particle bound transport Bioaccumulative Toxic to mammals, earthworms & honey bees Toxic to Beneficial insects (Parasitic wasps)-<i>Aphidius rhopalosiphi</i>, Beneficial insects (Predatory mites)-<i>Typhlodromus pyri</i> & Beneficial insects (Ground beetles)- <i>Poecilus cupreus</i> Highly toxic to aquatic organisms Reproductive / developmental toxicant Skin & eye irritant Skin sensitiser Possible liver and thyroid toxicant	Acute Tox. 4 (97.58%)	Skin Irrit. 2 (100%)	Skin Sens. 1 (100%)	Acute Tox. 4 (100%)	Repr. 1B (77.42%)	Repr. 1B (22.58%)	STOT RE 1 (88.71%)	STOT RE 2 (86.29%)	Aquatic Acute 1 (55.65%)	Aquatic Chronic 1 (100%)		H302+H332 (22.58%): Harmful if swallowed or if inhaled [Warning Acute toxicity, oral; acute toxicity, inhalation] H302 (97.58%): Harmful if swallowed [Warning Acute toxicity, oral] H315 (100%): Causes skin irritation [Warning Skin corrosion/irritation] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H332 (100%): Harmful if inhaled [Warning Acute toxicity, inhalation] H360 (77.42%): May damage fertility or the unborn child [Danger Reproductive toxicity] H360D (22.58%): May damage the unborn child [Danger Reproductive toxicity] H372 (88.71%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H373 (86.29%): May causes damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H400 (55.65%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4	Skin Irrit. 2	Skin Sens. 1	Acute Tox. 4	STOT RE 2	Aquatic Acute 1	Aquatic Chronic 1	Repr. 1B
	Mercury and compounds								Marine pollutant High solubility in water Highly volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable Non-persistent in soil Mobile in soil Low leachability Toxic to mammals, birds & earthworms Toxic to aquatic organisms Genotoxicity - Unspecified genotoxicity type Reproductive / developmental toxicant Skin & eye irritant Skin sensitiser Possible carcinogen & EDC Highly toxic US EPA - probable human carcinogen, CLP data - suspected carcinogen Possible liver toxicant	Met. Corr. 1 (15.73%)	Acute Tox. 2 (34.72%)	Acute Tox. 2 (98.55%)	Repr. 1B (94.76%)	STOT RE 1 (100%)	Aquatic Acute 1 (97.29%)	Aquatic Chronic 1 (100%)		H290 (15.73%): May be corrosive to metals [Warning Corrosive to Metals] H300+H330 (10.31%): Fatal if swallowed or if inhaled [Danger Acute toxicity, oral; acute toxicity, inhalation] H300 (34.72%): Fatal if swallowed [Danger Acute toxicity, oral] H330 (98.55%): Fatal if inhaled [Danger Acute toxicity, inhalation] H360 (94.76%): May damage fertility or the unborn child [Danger Reproductive toxicity] H372 (100%): Causes damage to organs through prolonged or repeated exposure [Danger Specific target organ toxicity, repeated exposure] H400 (97.29%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2	STOT RE 1	Aquatic Acute 1	Aquatic Chronic 1	Repr. 1B						

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[illegible]

16752-77-5	XX	Methomyl	1							Marine Pollutant subject to PIC regulations - CANDIDATE CHEMICAL Highly soluble in water Low volatility. If applied directly to plants, drift is a concern & mitigation is advisable From plant surface-Volatilisation is potentially a concern for short and long range air transport From soil surface-Volatilisation is not considered critical for air pollution Non-persistent in soil & water Mobile in soil Highly toxic to mammals, birds, honey bees & other bee species - <i>Trigona spinipes</i> & <i>Nannotrigona perilampoides</i> Toxic to earthworms & bumble bees Toxic to Beneficial insects (Ladybirds)-<i>Coccinella septempunctata</i> Adult, Beneficial insects (Lacewings)-<i>Chrysoperla carnea</i> Larva, Beneficial insects (Parasitic wasps)-<i>Aphidius rhopalosiphii</i> Adult & Beneficial insects (Predatory mites)-<i>Typhlodromus pyri</i> Protonymph Highly toxic to aquatic invertebrates, aquatic crustaceans & sediment dwelling organisms Toxic to fish Weak increase of aromatase activity and estrogen production Acetyl cholinesterase inhibitor Respiratory tract & eye irritant Possible thyroid toxicant	Acute Tox. 2 (100%) Acute Tox. 2 (30.82%) STOT SE 3 (30.82%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300+H330 (20.55%): Fatal if swallowed or if inhaled [Danger Acute toxicity, oral; acute toxicity, inhalation] H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H330 (30.82%): Fatal if inhaled [Danger Acute toxicity, inhalation] H336 (30.82%): May cause drowsiness or dizziness [Warning Specific target organ toxicity, single exposure; Narcotic effects] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 Aquatic Acute 1 Aquatic Chronic 1
74-83-9	XX	Methyl bromide					1			Montreal Protocol agreed phase out Marine Pollutant Chemical subject to PIC regulations- CHEMICALS RECOMMENDED FOR LISTING Highly volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable From soil surface-Volatilisation is potentially a concern for short and long range air transport Moderately persistent in soil Non-persistent in water Mobile in soil Low leachability Toxic to mammals & honeybees Highly toxic to birds Toxic to aquatic organisms Genotoxic-Unspecified genotoxicity type Possible Reproductive / developmental toxicant Neurotoxicant Respiratory tract, skin & eye irritant Mutagenic potential- Suspected of causing genetic defects Highly toxic Brain, kidney, respiratory toxicant	Flam. Gas 2 (45.81%) Press. Gas (Comp.) (87.74%) Acute Tox. 3 (100%) Skin Irrit. 2 (100%) Eye Irrit. 2 (100%) Acute Tox. 2 (45.81%) Acute Tox. 3 (54.19%) STOT SE 3 (100%) Muta. 2 (100%) STOT RE 2 (100%) Aquatic Acute 1 (100%) Ozone 1 (41.29%)		H221 (45.81%): Flammable gas [Danger Flammable gases] H280 (87.74%): Contains gas under pressure; may explode if heated [Warning Gases under pressure] H301+H331 (24.52%): Toxic if swallowed or if inhaled [Danger Acute toxicity, oral; acute toxicity, inhalation] H301 (100%): Toxic if swallowed [Danger Acute toxicity, oral] H315 (100%): Causes skin irritation [Warning Skin corrosion/irritation] H319 (100%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H330 (45.81%): Fatal if inhaled [Danger Acute toxicity, inhalation] H331 (54.19%): Toxic if inhaled [Danger Acute toxicity, inhalation] H335 (100%): May cause respiratory irritation [Warning Specific target organ toxicity, single exposure; Respiratory tract irritation] H341 (100%): Suspected of causing genetic defects [Warning Germ cell mutagenicity] H373 (100%): May causes damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H420 (41.29%): Harms public health and the environment by destroying ozone in the upper atmosphere [Warning Hazardous to the ozone layer]	Press. Gas Acute Tox. 3 Skin Irrit. 2 Acute Tox. 3 STOT SE 3 Muta. 2 STOT RE 2 Aquatic Acute 1 Ozone 1
21087-64-9	XX	Metribuzin					1			Potential groundwater contaminant Highly soluble in water Low volatility. If applied directly to plants, drift is a concern & mitigation is advisable Non-persistent in soils & water Mobile in soils High leachability Toxic to mammals, birds, earthworms, honey bees Toxic to Beneficial insects (Ladybirds)-<i>Coccinella septempunctata</i>, Beneficial insects (Lacewings)-<i>Chrysoperla carnea</i>, Beneficial insects (Parasitic wasps)-<i>Aphidius rhopalosiphii</i> & Beneficial insects (Predatory mites)-<i>Typhlodromus pyri</i> Toxic to aquatic organisms EDC -Endocrine issues - Hyperthyroidism, alteration of somatotropin levels Potential reproductive / developmental toxicant Thyroid toxicant	Acute Tox. 4 (100%) Acute Tox. 3 (25.2%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H331 (25.2%): Toxic if inhaled [Danger Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 H302 STOT RE 2 H373 (blood system) Aquatic Acute 1 H400 Aquatic Chronic 1 H410
7786-34-7	XX	Mevinphos	1						1	Severe Marine Pollutant Highly soluble in water Highly volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable Non-persistent in soil & water Mobile in soil Low leachability Highly toxic to mammals, birds, honey bees & other bee species - <i>Megachile rotundata</i> & <i>Nomia melanderi</i> Moderately harmful to Beneficial insects (Lacewings)- <i>Chrysoperla carnea</i> Harmful to Beneficial insects (Predatory mites)-<i>Typhlodromus pyri</i> Highly toxic to aquatic organisms Occupational Exposure Routes-Absorbed by the skin, inhalation and via the gastrointestinal tract - PPE/PPC essential Acetyl cholinesterase inhibitor Neurotoxicant Respiratory tract, skin & eye irritant	Acute Tox. 2 (100%) Acute Tox. 1 (100%) Acute Tox. 1 (12.77%) Acute Tox. 3 (80.85%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300+H310 (80.85%): Fatal if swallowed or in contact with skin [Danger Acute toxicity, oral; acute toxicity, dermal] H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H310 (100%): Fatal in contact with skin [Danger Acute toxicity, dermal] H330 (12.77%): Fatal if inhaled [Danger Acute toxicity, inhalation] H331 (80.85%): Toxic if inhaled [Danger Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 Acute Tox. 1 Aquatic Acute 1 Aquatic Chronic 1

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116714-46-6	XXY	N-methyl-2-pyrrolidone (Novaluron)								1	1	CO-FORMULANT Highly soluble in water Highly volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable Non-persistent in soil & water Mobile in soil Potential leachability Very toxic to mammals, birds, aquatic invertebrates & honeybees Toxic to earthworms & fish Occupational Exposure Routes - Occupational exposure may occur through inhalation and dermal contact Genotoxic - Genome mutation (EFSA), Gene mutation (EFSA) EDC Acetyl cholinesterase inhibitor Neurotoxicant Skin irritant Skin sensitiser Toxic if swallowed May cause cyanosis	Skin Irrit. 2 (100%) Eye Irrit. 2 (99.96%) STOT SE 3 (100%) Repr. 1B (96.31%)		H315 (100%): Causes skin irritation [Warning Skin corrosion/irritation] H319 (99.96%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H335 (100%): May cause respiratory irritation [Warning Specific target organ toxicity, single exposure; Respiratory tract irritation] H360 (96.31%): May damage fertility or the unborn child [Danger Reproductive toxicity]	Skin Irrit. 2 STOT SE 3 Eye Irrit. 2 STOT SE 3 Repr. 1B
	XY	Nonylphenol and Nonylphenol ethoxylates (NPEs)									1	CO-FORMULANT/SURFACTANT Highly volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable Non-persistent in water High leachability Highly toxic to mammals Toxic to aquatic organisms EDC Eye irritant Possible Skin irritant May disrupt hormone system Harmful if ingested May cause dermatitis	Acute Tox. 4 (100%) Skin Irrit. 2 (100%) Eye Irrit. 2 (100%) STOT RE 2 (100%) Aquatic Chronic 2 (100%)		H312 (100%): Harmful in contact with skin [Warning Acute toxicity, dermal] H315 (100%): Causes skin irritation [Warning Skin corrosion/irritation] H319 (100%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H373 (100%): May cause damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H411 (100%): Toxic to aquatic life with long lasting effects [Hazardous to the aquatic environment, long-term hazard]	
1113-02-6	XX	Omethoate								1	1	Potential groundwater contaminant Groundwater monitoring programmes have identified it as a pollutant Low aqueous solubility Not volatile Persistent in soil Not persistent in water systems Slightly mobile in soil Potential for particle bound transport Bioaccumulates Toxic to birds, honeybees & earthworms Toxic to aquatic organisms Public Exposure Routes - small risk of damage/repair (EFSA) Skin sensitiser Possible carcinogen & skin irritant Potential liver, kidney & spleen toxicant	Acute Tox. 2 (92.68%) Acute Tox. 3 (92.68%) Aquatic Acute 1 (100%)		H300 (92.68%): Fatal if swallowed [Danger Acute toxicity, oral] H311 (92.68%): Toxic in contact with skin [Danger Acute toxicity, dermal] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard]	Acute Tox. 3 H301 Acute Tox. 4 H312 Aquatic Acute 1 H400
190044-88-3	XX	Oryzalin									1	Low aqueous solubility Low volatility. If applied directly to plants, drift is a concern & mitigation is advisable Persistent in soil & water Slightly mobile in soil High potential for particle bound transport Bioaccumulates Toxic to earthworms Toxic to aquatic organisms Potential Reproductive / developmental toxicant Possible carcinogen Possible liver toxicant Respiratory tract irritant May cause anaemia US EPA - possible human carcinogen	Skin Sens. 1 (98.37%) Aquatic Acute 1 (98.37%) Aquatic Chronic 1 (98.37%)		H317 (98.37%): May cause an allergic skin reaction [Warning Sensitization, Skin] H400 (98.37%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (98.37%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Skin Sens. 1 H317 Aquatic Acute 1 H400 Aquatic Chronic 1 H410 Aquatic Chronic 4 H413 Aquatic Chronic 2 H411 Skin Sens. 1 H317 Carc. 2 H351 (Suspected of ca...) STOT RE 2 H373 (otherBlood, KI...) (May cause damag...)
19666-30-9	XX	Oxadiazon									1	Marine Pollutant Highly soluble in water Low volatility Not persistent in soil or water systems Very mobile on soil Highly toxic to mammals, birds & honeybees Toxic to earthworms & bumble bees Toxic to aquatic organisms EDC - Endocrine issues - weak estrogen effect Acetyl cholinesterase inhibitor Neurotoxicant Skin irritant Harmful if ingested	Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Aquatic Acute 1 H400 Aquatic Chronic 1 H410

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23135-22-0	XX	Oxamyl	1							Marine Pollutant water contaminant Low volatility & water Highly toxic to mammals, birds, honey bees & bumble bees Toxic to earthworms & Beneficial insects (Lacewings) - <i>Chrysoperla carnea</i> Larva, Parasitic wasps-<i>Aphidius rhopalosiph</i>, Predatory mites - <i>Typhlodromus pyri</i> & Ground beetles-<i>Poecilus cupreus</i> Toxic to aquatic organisms issues - weak estrogen effect Neurotoxicant inhibitor Potential Eye irritant High solubility in water Non-persistent in soil Very mobile in soil EDC - Endocrine Acetyl cholinesterase Harmful if ingested Skin irritant	Acute toxicity (inhalation) - category 2 Acute toxicity (ingestion) - category 2 Acute toxicity (dermal) - category 4 Hazardous to the aquatic environment (chronic) - category 2		H300: Fatal if swallowed [Danger Acute toxicity, oral] H312: Harmful in contact with skin [Warning Acute toxicity, dermal] H330: Fatal if inhaled [Danger Acute toxicity, inhalation] H411: Toxic to aquatic life with long lasting effects [Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 H300 Acute Tox. 4 H312 Acute Tox. 2 H330 Aquatic Chronic 2 H411
		Oxathiapiprolin								Low solubility in water volatility soil Low leachability Toxic to birds & honeybees Beneficial insects (Lacewings)-<i>Chrysoperla carnea</i>, Parasitic wasps-<i>Aphidius rhopalosiph</i> & Predatory mites-<i>Typhlodromus pyri</i> Toxic to aquatic organisms Possible Reproductive / developmental toxicant Possible liver & gall bladder toxicant Low Moderately persistent in Non-mobile in soil Toxic to	Aquatic Acute 1 (70.83%) Aquatic Chronic 1 (100%)		H400 (70.83%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Aquatic Chronic 1 H410
		Oxyfluorfen								Possible ground water pollutant solubility in water Moderately persistent in soil in soil High potential for particle bound transport Toxic to birds & earthworms Toxic to fish, aquatic invertebrates, sediment dwelling organisms & algae aquatic plants Routes-Acceptable risk to operator for intended uses Risks identified for workers re-entering sprayed ruit/wine areas - cloed re-entry periods must be imposed Genotoxic - Unspecified genotoxicity type Possible Carcinogen & eye irritant toxicant US EPA - possible human carcinogen suspected carcinogen Low Non-mobile Low leachability Bioaccumulative Highly toxic to Occupational Exposure Possible Skin Liver and spleen CLP data -	Aquatic Acute 1 (98.95%) Aquatic Chronic 1 (98.95%)		H400 (98.95%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (98.95%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Aquatic Acute 1 H400 Aquatic Chronic 1 H410 Skin Irrit. 2 H315 Eye Irrit. 2 H319 STOT SE 3 H335 (other:respirato...) Carc. 2 H351 Acute Tox. 4 H302 Acute Tox. 4 H312 Acute Tox. 4 H332 Carc. 2 H351
		Paclobutrazol								Persistent-Bioaccumulative-Toxic pollutant Non-volatile volatility persistent in soil mobile in soil Toxic to mammals, earthworms & honeybees Toxic to aquatic organisms Highly toxic to aquatic plants eye irritant toxicant swallowed Low water solubility Low Non-Moderately Respiratory tract, skin & eye Possible Reproductive / developmental Harmful if inhaled or	Flam. Sol. 1 (49.57%) Acute Tox. 4 (100%) (47.83%) Acute Tox. 4 (47.83%) Skin Irrit. 2 (13.91%) Skin Sens. 1 (13.91%) Eye Irrit. 2 (45.22%) Acute Tox. 2 (33.91%) STOT SE 3 (13.91%) Repr. 2 (19.13%) Repr. 2 (26.09%) Aquatic Acute 1 (44.35%) Aquatic Chronic 1 (44.35%)		H228 (49.57%): Flammable solid [Danger Flammable solids] H302+H332 (24.35%): Harmful if swallowed or if inhaled [Warning Acute toxicity, oral; acute toxicity, inhalation] H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H312 (47.83%): Harmful in contact with skin [Warning Acute toxicity, dermal] H315 (13.91%): Causes skin irritation [Warning Skin corrosion/irritation] H317 (13.91%): May cause an allergic skin reaction [Warning Sensitization, Skin] H319 (45.22%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H332 (33.91%): Harmful if inhaled [Warning Acute toxicity, inhalation] H335 (13.91%): May cause respiratory irritation [Warning Specific target organ toxicity, single exposure; Respiratory tract irritation] H361 (19.13%): Suspected of damaging fertility or the unborn child [Warning Reproductive toxicity] H361d (26.09%): Suspected of damaging the unborn child [Warning Reproductive toxicity] H400 (44.35%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (44.35%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 H302 Eye Irrit. 2 H319 Acute Tox. 4 H332 Aquatic Acute 1 H400 Aquatic Chronic 1 H410 Repr. 2 H361d

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52645-53-1	XX	Permethrin								Low solubility in water Moderately persistent in soil & water Low leachability High potential for particle bound transport Bioaccumulative Highly toxic to honey bees, bumble bees, mason bees & other bee species - <i>Megachile rotundata</i> & <i>Trigona spinipes</i> Toxic to mammals Insects (Ladybirds)- <i>Coccinella septempunctata</i> Adult & Lacewings- <i>Chrysoperla carnea</i> Larva Chrysoperla & Predatory mites - <i>Typhlodromus pyri</i> Highly toxic to aquatic organisms EDC-Endocrine issues - Inhibition of estrogen-sensitive cells proliferation Reproductive / developmental toxicant Neurotoxicant Skin sensitiser cholinesterase inhibitor Estrogenic	Low volatility Non-mobile in soil High potential for particle bound transport Toxic to Beneficial insects (Ladybirds)- <i>Coccinella septempunctata</i> Adult & Lacewings- <i>Chrysoperla carnea</i> Larva Harmful to Parasitic wasps & Predatory mites - <i>Typhlodromus pyri</i> Toxic to algae Skin & eye irritant Possible Acetylcholinesterase inhibitor Harmful	Acute Tox. 4 (100%) Skin Sens. 1 (88.85%) Acute Tox. 4 (100%) Aquatic Acute 1 (99.81%) Aquatic Chronic 1 (100%)		H302+H332 (11.8%): Harmful if swallowed or if inhaled [Warning Acute toxicity, oral; acute toxicity, inhalation]H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H317 (88.85%): May cause an allergic skin reaction [Warning Sensitization, Skin] H332 (100%): Harmful if inhaled [Warning Acute toxicity, inhalation] H400 (99.81%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 Skin Sens. 1 H302 Acute Tox. 4 H332 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
	Y	Petrolatum								CO-FORMULANT Persistent in the environment (water, soil, air) Secondary poisoning risks for predators (PNEC oral 9.33 mg/kg food) Inhalation & dermal risk Reproductive / developmental toxicant Carcinogenic. Genotoxic (tetragenocyt) (obtained from ECHA CHEM)		Carc. 1B (96.57%)		H350 (96.57%): May cause cancer [Danger Carcinogenicity]	Carc. 1B H350
298-02-2	XX	Phorate								Severe Marine Pollutant subject to PIC regulations - LISTED UNDER ANNEX III Moderate water solubility If applied directly to plants or soil, drift is a concern & mitigation is advisable Non-persistent in water Slightly mobile in soil Bioaccumulates mammals, birds & honeybees Highly toxic to aquatic organisms Occupational Exposure Routes - May be absorbed through the skin Highly toxic by skin contact Acetyl cholinesterase inhibitor Neurotoxicant Skin & eye irritant Very toxic	Chemical Highly volatile. Moderately persistent in soil Low leachability Highly toxic to earthworms & bumble bees Toxic to earthworms & bumble bees Highly toxic to aquatic organisms Acetyl cholinesterase inhibitor Skin & eye irritant Very toxic	Acute Tox. 2 (100%) Acute Tox. 1 (100%) Acute Tox. 1 (66.67%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300+H310+H330 (66.67%): Fatal if swallowed, in contact with skin or if inhaled [Danger Acute toxicity, oral; acute toxicity, dermal; acute toxicity, inhalation] H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H310 (100%): Fatal in contact with skin [Danger Acute toxicity, dermal] H330 (66.67%): Fatal if inhaled [Danger Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 H300 Acute Tox. 1 H310 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
13171-21-6	XXX	Phosphamidon								Severe Marine Pollutant Convention (Class Ia) - some formulations subject to PIC regulations -- ANNEX III High solubility in water Low volatility. If applied directly to plants or soil, drift is a concern & mitigation is advisable Non-persistent in soil Moderately persistent in water Mobile in soil Highly toxic to mammals, birds, honeybees & other bee species - <i>Megachile rotundata</i> & <i>Nomia melanderi</i> Highly toxic to aquatic invertebrates & aquatic crustaceans Toxic to fish Occupational Exposure Routes - Readily absorbed from the gastrointestinal tract, through the skin & by inhalation of spray mists & dusts Genotoxic EDC - Mutagenic potential; Suspected of causing genetic defects Acetyl cholinesterase inhibitor Neurotoxicant Possible carcinogenic Extremely hazardous US EPA - possible human carcinogen	Rotterdam High solubility in water Non-persistent in soil Moderately persistent in water Mobile in soil Highly toxic to mammals, birds, honeybees & other bee species - <i>Megachile rotundata</i> & <i>Nomia melanderi</i> Highly toxic to aquatic invertebrates & aquatic crustaceans Toxic to fish Occupational Exposure Routes - Readily absorbed from the gastrointestinal tract, through the skin & by inhalation of spray mists & dusts Genotoxic EDC - Mutagenic potential; Suspected of causing genetic defects Acetyl cholinesterase inhibitor Neurotoxicant Possible carcinogenic Extremely hazardous US EPA - possible human carcinogen	Acute Tox. 2 (100%) Acute Tox. 3 (87.23%) Acute Tox. 1 (12.77%) Muta. 2 (87.23%) Aquatic Acute 1 (87.23%) Aquatic Chronic 1 (87.23%)		H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H311 (87.23%): Toxic in contact with skin [Danger Acute toxicity, dermal] H330 (12.77%): Fatal if inhaled [Danger Acute toxicity, inhalation] H341 (87.23%): Suspected of causing genetic defects [Warning Germ cell mutagenicity] H400 (87.23%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (87.23%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 H300 Acute Tox. 3 H311 Muta. 2 H341 Aquatic Acute 1 H400 Aquatic Chronic 1 H410

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		Propaquizafop								Low water solubility Moderately persistent in soil & water Medium potential for particle bound transport Bioaccumulates honeybees & beneficial insects-Ladybirds-<i>Coccinella septempunctata</i> corr, Lacewings-<i>Chrysoperla carnea</i> corr, Parasitic wasps-Aphidius <i>rhopalosiph</i>, Predatory mites-Typhlodromus <i>pyri</i> & Ground beetles-<i>Poecilus cupreus</i> corr Toxic to earthworms, Toxic to Beneficial insects (Parasitic wasps)-Aphidius <i>rhopalosiph</i> & Predatory mites-Typhlodromus <i>pyri</i> Genotoxic (chromosome aberration) (EFSA) Possible carcinogen and reproductive & developmental toxicant Potential skin irritant	Low volatility Mobile in soils Acute Tox. 4 (55.88%) Aquatic Acute 1 (44.12%) Aquatic Chronic 1 (44.12%)		H317 (44.12%): May cause an allergic skin reaction [Warning Sensitization, Skin] H332 (55.88%): Harmful if inhaled [Warning Acute toxicity, inhalation] H400 (44.12%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (44.12%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 H332 Skin Sens. 1 H317 Aquatic Acute 1H400 Aquatic Chronic 1 H410
2312-35-8	XX	Propargite							Low solubility in water Non-persistent in soil Low leachability Toxic to earthworms & honeybees Insects (Parasitic wasps)-Aphidius <i>rhopalosiph</i> & Predatory mites-Typhlodromus <i>pyri</i> aquatic organisms developmental toxicant Skin sensitiser Toxic via inhalation May cause dermatitis US EPA - probable human carcinogen	Low volatility Moderately Non-mobile in soil Bioaccumulates Toxic to Beneficial insects (Parasitic wasps)-Aphidius <i>rhopalosiph</i> & Predatory mites-Typhlodromus <i>pyri</i> Highly toxic to aquatic organisms Public Exposure Routes - Small risk to bystanders identified Reproductive / Skin & eye irritant Possible carcinogen	Skin Irrit. 2 (100%) Eye Dam. 1 (100%) Acute Tox. 3 (94.64%) Carc. 2 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H315 (100%): Causes skin irritation [Warning Skin corrosion/irritation] H318 (100%): Causes serious eye damage [Danger Serious eye damage/eye irritation] H331 (94.64%): Toxic if inhaled [Danger Acute toxicity, inhalation] H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Skin Irrit. 2 H315 Eye Dam. 1 H318 Acute Tox. 3 H331 Carc. 2 H351 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
122-42-9	XXX	Propham							Moderate solubility in soil applied directly to plants or soil, drift is a concern & mitigation is advisable Very persistent in water soil Toxic to honeybees Toxic to aquatic organisms cholinesterase inhibitor & neurotoxicant Harmful if swallowed	Highly volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable Non-persistent in soil Moderately mobile in soil Possible Acetyl	Acute Tox. 4 (100%)		H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral]	Acute Tox. 4 H302
60207-90-1	XX	Propiconazole						1	Moderate solubility in water Volatilisation is potentially a concern for short and long range air transport Moderately persistent in soils Moderately persistent in water mobile particle bound transport index Toxic to mammals, earthworms & mason bees Toxic to aquatic organisms Endocrine issues - Weak estrogen and aromatase activity inhibition Reproductive / developmental toxicant Respiratory tract irritant carcinogen, skin irritant & sensitizer US EPA - possible human carcinogen Possible liver toxicant	Low volatility Low volatility Slightly Potential for Bioaccumulates EDC - Possible	Acute Tox. 4 (100%) Skin Sens. 1 (100%) Repr. 1A (13.45%) Repr. 1B (35.67%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H360 (13.45%): May damage fertility or the unborn child [Danger Reproductive toxicity] H360D (35.67%): May damage the unborn child [Danger Reproductive toxicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 H302 Skin Sens. 1 H317 Aquatic Acute 1 H400 Aquatic Chronic 1 Repr. 1B H360D
9016-72-2	XX	Propineb						1	Low solubility in water applied directly to plants, drift is a concern & mitigation is advisable Non-persistent in soil & water Toxic to earthworms (Parasitic wasps)-Aphidius <i>rhopalosiph</i> & Predatory mites-Typhlodromus <i>pyri</i> organisms Possible carcinogen, EDC, reproductive /developmental toxicant, respiratory tract & skin irritant, skin sensitiser Primarily distributes via the thyroid gland, potentially thyroid toxicant May damage lungs or cause muscular problems US EPA - probable human carcinogen	Low volatility. If applied directly to plants or soil, drift is a concern & mitigation is advisable Toxic to Beneficial insects (Parasitic wasps)-Aphidius <i>rhopalosiph</i> & Predatory mites-Typhlodromus <i>pyri</i> Toxic to aquatic organisms Possible carcinogen, EDC, reproductive /developmental toxicant, respiratory tract & skin irritant, skin sensitiser Primarily distributes via the thyroid gland, potentially thyroid toxicant May damage lungs or cause muscular problems US EPA - probable human carcinogen	Not classified under GHS		Not classified under GHS	Acute Tox. 4 H312 Skin Sens. 1 H317 STOT RE 2 H373 (oral) Aquatic Acute 1 H400 Aquatic Chronic 2 H411
114-26-1	XX	Propoxur						1	Marine Pollutant Mobile in groundwaters Not highly volatile Low volatility. If applied directly to plants or soil, drift is a concern & mitigation is advisable Persistent in soil systems Persistent in water Highly toxic to mammals, birds & honeybees Toxic to bumble bees Harmful to Beneficial insects (Lacewings)- <i>Chrysoperla carnea</i> Toxic to aquatic organisms Highly toxic to humans if it is ingested EDC - Endocrine issues - Weak estrogenic effect Potential Carcinogen, Acetyl cholinesterase inhibitor, Neurotoxicant & Eye irritant Toxic US EPA - probable human carcinogen	High aqueous solubility Mobile in soil High leachability Non Persistent in water Toxic to bumble bees <i>Chrysoperla carnea</i> Highly toxic to humans if it is ingested EDC - Endocrine issues - Weak estrogenic effect Potential Carcinogen, Acetyl cholinesterase inhibitor, Neurotoxicant & Eye irritant Toxic US EPA - probable human carcinogen	Acute Tox. 2 (23.31%) Acute Tox. 3 (44.79%) Acute Tox. 3 (32.52%) Acute Tox. 3 (22.7%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (68.1%)		H300 (23.31%): Fatal if swallowed [Danger Acute toxicity, oral] H301 (44.79%): Toxic if swallowed [Danger Acute toxicity, oral] H311 (32.52%): Toxic in contact with skin [Danger Acute toxicity, dermal] H331 (22.7%): Toxic if inhaled [Danger Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (68.1%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 H301 Aquatic Acute 1 H400 Aquatic Chronic 1 H410

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123312-89-0	XX	Pymetrozine								Moderate solubility in water Non-persistent in soil Low leachability Medium potential for particle bound transport Toxic to earthworms, honey bees & bumble bees Toxic to aquatic organisms developmental toxicant Possible carcinogen Highly toxic - harmful if swallowed US EPA - likely to be human carcinogen CLP data - suspected carcinogen	Low volatility Slightly mobile	Acute Tox. 4 (21.47%) Carc. 2 (100%) Repr. 2 (74.01%) Aquatic Chronic 1 (75.14%) Aquatic Chronic 3 (24.86%)		H332 (21.47%): Harmful if inhaled [Warning Acute toxicity, inhalation] H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H361fd (74.01%): Suspected of damaging fertility; Suspected of damaging the unborn child [Warning Reproductive toxicity] H410 (75.14%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard] H412 (24.86%): Harmful to aquatic life with long lasting effects [Hazardous to the aquatic environment, long-term hazard]	Carc. 2 H351 Aquatic Chronic 1 H410 Repr. 2 H361fd
13593-03-8	XX	Quinalphos							1	Rotterdam Convention (Class Ia) - subject to PIC regulations -- CANDIDATE CHEMICAL Marine Pollutant Low water solubility Non volatile Moderately persistent in water Low leachability mammals, birds, honeybees & bumble bees Highly toxic to aquatic organisms cholinesterase inhibitor Neurotoxicant Respiratory tract, skin & eye irritant & reproductive / developmental toxicant Highly toxic, may be fatal if inhaled, swallowed or absorbed through skin	Low volatility. If applied directly to plants, drift is a concern & mitigation is advisable Non-persistent in soil Slightly mobile in soils Highly toxic to Acetyl Possible carcinogen	Acute Tox. 2 (12.77%) Acute Tox. 3 (87.23%) Acute Tox. 3 (12.77%) Acute Tox. 4 (87.23%) Eye Dam. 1 (80.85%) Eye Irrit. 2 (12.77%) Acute Tox. 1 (80.85%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300 (12.77%): Fatal if swallowed [Danger Acute toxicity, oral] H301 (87.23%): Toxic if swallowed [Danger Acute toxicity, oral] H311 (12.77%): Toxic in contact with skin [Danger Acute toxicity, dermal] H312 (87.23%): Harmful in contact with skin [Warning Acute toxicity, dermal] H318 (80.85%): Causes serious eye damage [Danger Serious eye damage/eye irritation] H319 (12.77%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H330 (80.85%): Fatal if inhaled [Danger Acute toxicity, inhalation] H351 (12.77%): Suspected of causing cancer [Warning Carcinogenicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 H301 Acute Tox. 4 H312 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
124495-18-7	XX	Quinoxifene							1	Low solubility in water Persistent in soil High potential for particle bound transport Low leachability Toxic to earthworms Possible skin sensitiser and blood toxicant	Low volatility Non-mobile in soil Bioaccumulates Toxic to aquatic organisms Possible skin & eye irritant Possible liver, kidney	Skin Sens. 1 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Skin Sens. 1 H317 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
		Quizalofop-P-tefuryl							1	Severe Marine Pollutant Water solubility Non-persistent in soils & water earthworms Genotoxic - Unspecified genotoxicity type / developmental toxicant Eye irritant Liver toxicant CLP data - suspected carcinogen Moderately toxic Liver toxicant CLP data - suspected carcinogen	Low volatility Moderately mobile Toxic to mammals & earthworms Toxic to aquatic organisms Reproductive / development Skin sensitiser Possible Moderately toxic	Acute Tox. 4 (100%) Muta. 2 (100%) Repr. 1B (100%) STOT RE 2 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H341 (100%): Suspected of causing genetic defects [Warning Germ cell mutagenicity] H360 (100%): May damage fertility or the unborn child [Danger Reproductive toxicity] H373 (100%): May cause damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 Muta. 2 Repr. 1B STOT RE 2 Aquatic Acute 1 Aquatic Chronic 1
122-34-9	XX	Simazine							1	Potential groundwater & surface water contaminant Low solubility in water Moderately persistent in soil & water transport Toxic to earthworms & honeybees organisms Possible carcinogen May cause tremor, convulsions, paralysis, cyanosis, slowed respiration and diarrhoea CLP data - suspected carcinogen; US EPA - possible human carcinogen	Low volatility Moderately mobile in soil Medium potential for particle bound transport Bioaccumulates Toxic to aquatic organisms EDC - Endocrine issues - Induction of aromatase activity, increase of estrogen production May cause dermatitis CLP data - suspected carcinogen	Acute Tox. 4 (52.38%) Eye Irrit. 2 (53.57%) Carc. 2 (100%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H302 (52.38%): Harmful if swallowed [Warning Acute toxicity, oral] H319 (53.57%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Carc. 2 Aquatic Acute 1 Aquatic Chronic 1

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		Tefluthrin								Low solubility in water applied directly to plants or soil, drift is a concern & mitigation is advisable Non-persistent in soil Non-mobile in soil Medium potential for particle bound transport Bioaccumulative mammals, earthworms & honey bees Toxic to birds aquatic invertebrates Skin & eye irritant Reproductive / developmental toxicant Possible thyroid and CNS toxicant Moderately volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable Non-persistent in soil Low leachability Highly toxic to mammals, earthworms & honey bees Highly toxic to fish & aquatic invertebrates Toxic to sediment dwelling organisms & algae Neurotoxicant Possible EDC & Highly toxic	Acute Tox. 2 (100%) Acute Tox. 2 (98.5%) Acute Tox. 1 (81%) Acute Tox. 3 (19%) Aquatic Acute 1 (98.5%) Aquatic Chronic 1 (79.5%)		H300+H310+H330 (14%): Fatal if swallowed, in contact with skin or if inhaled [Danger Acute toxicity, oral; acute toxicity, dermal; acute toxicity, inhalation] H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H310 (98.5%): Fatal in contact with skin [Danger Acute toxicity, dermal] H330 (81%): Fatal if inhaled [Danger Acute toxicity, inhalation] H331 (19%): Toxic if inhaled [Danger Acute toxicity, inhalation] H400 (98.5%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (79.5%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 H300 Acute Tox. 2 H310 Acute Tox. 1 H330 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
		Terbufos	1						1	Severe Marine Pollutant PIC regulations - LISTED UNDER ANNEX III Found infrequently in drinking water supplies Low solubility in water applied directly to plants or soil, drift is a concern & mitigation is advisable Non-persistent in soil & water Slightly mobile in soil Low leachability Bioaccumulates Highly toxic to mammals & earthworms Toxic to birds & honeybees except algae Neurotoxicant Highly toxic Chemical subject to Highly volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable Non-persistent in soil Bioaccumulates Toxic to birds & honeybees Toxic to algae Acetyl cholinesterase inhibitor Skin & eye irritant	Acute Tox. 2 (100%) Acute Tox. 1 (100%) Acute Tox. 1 (37.33%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H300+H310+H330 (37.33%): Fatal if swallowed, in contact with skin or if inhaled [Danger Acute toxicity, oral; acute toxicity, dermal; acute toxicity, inhalation] H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral] H310 (100%): Fatal in contact with skin [Danger Acute toxicity, dermal] H330 (37.33%): Fatal if inhaled [Danger Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 2 H300 Acute Tox. 1 H310 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
13071-79-9	XX	Tetraconazole	1						1	Common environmental pollutant Insoluble in water Persistent in both soils and water Low leachability High potential for particle bound transport Toxic to mammals Highly toxic to aquatic organisms humans through ingestion, inhalation & via skin absorption Public Exposure Routes - Risk from residues in food Occupational Exposure Routes - PPE/PPC essential Genotoxic EDC - Endocrine issues - Competitive binding to androgen receptors Reproductive / developmental toxicant & eye irritant Possible carcinogen Moderately toxic by ingestion, inhalation and skin absorption Metabolite of DDT Not highly volatile Non-mobile in soils High potential for particle bound transport Toxic to mammals Toxic to humans through ingestion, inhalation & via skin absorption Respiratory tract, skin & eye irritant	Acute Tox. 4 (100%) Acute Tox. 4 (100%) Aquatic Acute 1 (21.37%) Aquatic Chronic 2 (100%)		H302+H332 (88.55%): Harmful if swallowed or if inhaled [Warning Acute toxicity, oral; acute toxicity, inhalation] H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H332 (100%): Harmful if inhaled [Warning Acute toxicity, inhalation] H400 (21.37%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (21.37%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard] H411 (100%): Toxic to aquatic life with long lasting effects [Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 Acute Tox. 4 Aquatic Chronic 2
		Tetrahydrofurfuryl alcohol							1	CO-FORMULANT Causes serious eye irritation	Eye Irrit. 2A (100%)		H319 (100%): Causes serious eye irritation [Warning Serious eye damage/eye irritation]	Eye Irrit 2A H319
		Thiabendazole							1	Low solubility in water Very persistent in soil Slightly mobile in soil particle bound transport Toxic to aquatic organisms aberration (EFSA) developmental toxicant & Respiratory tract irritant Possible thyroid, liver, spleen and kidney toxicant Low volatility Persistent in water High potential for particle bound transport Toxic to earthworms & honeybees Genotoxic - Chromosome aberration (EFSA) Possible Carcinogen, Reproductive / developmental toxicant & Respiratory tract irritant	Aquatic Acute 1 (50.00%) Aquatic Chronic 1 (50.00%)		H400 (99.46%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (99.73%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Aquatic Acute 1 H400 Aquatic Chronic 1 H410

111988-49-9	XX	Thiacloprid									Moderate water solubility Non-persistent in soils mobile Low volatility Slightly Low leachability Toxic to mammals, earthworms, honey bees & mason bees Highly toxic to birds & other bee species - <i>Nannotrigona perilampoides</i> Highly toxic to aquatic crustaceans & sediment dwelling organisms EDC Reproductive / developmental toxicant Neurotoxicant eye irritant carcinogen thyroid toxicant US EPA - likely to be a human carcinogen; CLP data - suspected carcinogen	Acute Tox. 3 (64.81%) Acute Tox. 4 (35.19%) Acute Tox. 4 (100%) STOT SE 3 (64.81%) Carc. 2 (64.81%) Repr. 1B (38.89%) Repr. 1B (27.78%) Aquatic Acute 1 (64.81%) Aquatic Chronic 1 (64.81%)		H301 (64.81%): Toxic if swallowed [Danger Acute toxicity, oral] H302 (35.19%): Harmful if swallowed [Warning Acute toxicity, oral] H332 (100%): Harmful if inhaled [Warning Acute toxicity, inhalation] H336 (64.81%): May cause drowsiness or dizziness [Warning Specific target organ toxicity, single exposure; Narcotic effects] H351 (64.81%): Suspected of causing cancer [Warning Carcinogenicity] H360 (38.89%): May damage fertility or the unborn child [Danger Reproductive toxicity] H360FD (27.78%): May damage fertility; May damage the unborn child [Danger Reproductive toxicity] H400 (64.81%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (64.81%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 3 Acute Tox. 4 STOT SE 3 Carc. 2 Aquatic Acute 1 Aquatic Chronic 1 Repr. 1B
153719-23-4	XX	Thiamethoxam									High solubility in water Moderately persistent in soil High leachability birds Low volatility Mobile in soil Toxic to mammals & birds Highly toxic to honeybees, bumble bees & other bee species - <i>Trigona iridipennis</i> & <i>Nannotrigona perilampoides</i> Toxic to aquatic organisms Possible Carcinogen & Skin irritant Increased incidence of liver cell adenoma and adenocarcinoma in mice US EPA - likely to be human carcinogen	Fam. Sol. 2 (46.8%) Acute Tox. 4 (100%) Acute Tox. 4 (48.1%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (100%)		H228 (46.8%): Flammable solid [Danger Flammable solids] H302+H332 (48.1%): Harmful if swallowed or if inhaled [Warning Acute toxicity, oral; acute toxicity, inhalation] H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H332 (48.1%): Harmful if inhaled [Warning Acute toxicity, inhalation] H400 (100%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (100%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 Acute Tox. 4 H302 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
31895-21-3	XX	Thiocyclam									High water solubility applied directly to plants, drift is a concern & mitigation is advisable Non-persistent in soil & water Low leachability earthworms & honey bees Harmful to Beneficial insects (Lacewings)-<i>Chrysoperla carnea</i> & Predatory mites-<i>Typhlodromus pyri</i> Highly toxic to fish invertebrates & algae Skin sensitiser Low volatility. If mitigation is advisable Mobile in soil Toxic to mammals, Toxic to aquatic invertebrates & algae Skin & eye irritant Harmful if swallowed	Acute toxicity (Oral) - Category 3 Acute toxicity (Inhalation: Dusts and mists) - Category 4 Skin corrosion/irritation - Category 3 Specific target organ toxicity - Single exposure - Category 1 (nervous system) Specific target organ toxicity - Repeated exposure - Category 2 (nervous system) Hazardous to the aquatic environment (Acute) - Category 1 Hazardous to the aquatic environment (Long-term) - Category 1		H301: Toxic if swallowed [Danger Acute toxicity, oral] H316: Causes mild skin irritation [Warning Skin corrosion/irritation] H332: Harmful if inhaled [Warning Acute toxicity, inhalation] H370: Causes damage to organs [Danger Specific target organ toxicity, single exposure] H373: May causes damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H400: Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410: Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 Acute Tox. 4 H302 Aquatic Acute 1 H400 Aquatic Chronic 1 H410

[illegible]

UnPoison South Africa's Highly Hazardous Pesticide Database vs EU's According to JMPM Criteria (FAO/WHO), GHS and ECHA Classification Systems - February 2025

		Triclopyr								Potential groundwater pollutant Low volatility. If applied directly to plants, drift is a concern & mitigation is advisable Non-persistent in water High leachability Toxic to mammals, birds & earthworms Toxic to aquatic plants Occupational Exposure Routes-Risk to operators but acceptable under label recommendations for use for personal protection clothing and equipment Reproductive / developmental toxicant Skin sensitizer Highly toxic Heart, kidney and liver toxicant	High water solubility Moderately persistent in soil Mobile in soil Toxic to mammals, birds & earthworms Reproductive / developmental toxicant Eye irritant	Acute Tox. 4 (100%) Skin Sens. 1 (59.2%) Eye Irrit. 2 (60.2%) STOT RE 2 (54.4%) Aquatic Acute 1 (54.4%) Aquatic Chronic 1 (54.4%)		H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral] H317 (59.2%): May cause an allergic skin reaction [Warning Sensitization, Skin] H319 (60.2%): Causes serious eye irritation [Warning Serious eye damage/eye irritation] H373 (54.4%): May cause damage to organs through prolonged or repeated exposure [Warning Specific target organ toxicity, repeated exposure] H400 (54.4%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (54.4%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Acute Tox. 4 H302 Skin Sens. 1 H317 Eye Irrit. 2 H319 STOT RE 2 H373 (kidney) Aquatic Acute 1 H400 Aquatic Chronic 1 H410 Repr. 2 H361 Aquatic Chronic 4 H413 STOT SE 2 H371 (Organs)
64628-44-0	XX	Triflururon							1	Low solubility in water Non-persistent in soil Low leachability particle bound transport Bioaccumulates earthworms Toxic to algae irritant	Low volatility Slightly mobile n soil Medium potential for Toxic to birds & earthworms Highly toxic to aquatic invertebrates Respiratory tract	Acute Tox. 2 (55.9%) Aquatic Acute 1 (44.1%) Aquatic Chronic 1 (44.1%), 2 (95%)		H330 (95%): Fatal if inhaled [Danger Acute toxicity, inhalation] H400 (44.1%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (44.1%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Aquatic Acute 1 H400 Aquatic Chronic 1 H410 Acute Tox. 2 H330
1582-09-8	XX	Trifluralin							1	Low solubility in water Non-persistent in soil Low leachability Toxic to earthworms organisms EDC- Endocrine issues - Interaction with pregnane X cellular receptor Reproductive / developmental toxicant Skin sensitizer toxicant May cause respiratory depression human carcinogen	Moderately volatile. If applied directly to plants or soil, drift is a concern & mitigation is advisable From soil surface-Volatilisation is potentially a concern for short and long range air transport Non-mobile in soil Bioaccumulates Toxic to aquatic crustaceans Highly toxic to aquatic crustaceans Respiratory tract irritant Blood, liver and kidney US EPA - possible suspected carcinogen CLP data - suspected carcinogen	Acute Tox. 4 (25.42%) Skin Sens. 1 (100%) Acute Tox. 2 (24.29%) Carc. 2 (97.74%) Aquatic Acute 1 (98.87%) Aquatic Chronic 1 (99.44%)		H302 (25.42%): Harmful if swallowed [Warning Acute toxicity, oral] H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H330 (24.29%): Fatal if inhaled [Danger Acute toxicity, inhalation] H351 (97.74%): Suspected of causing cancer [Warning Carcinogenicity] H400 (98.87%): Very toxic to aquatic life [Warning Hazardous to the aquatic environment, acute hazard] H410 (99.44%): Very toxic to aquatic life with long lasting effects [Warning Hazardous to the aquatic environment, long-term hazard]	Skin Sens. 1 H317 Carc. 2 H351 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
83657-22-1	XX	Uniconazole							1	PLANT GROWTH REGULATOR contaminant Very persistent in soil & aquatic systems Toxic to honey bees, birds & earthworms aquatic organisms Possible skin & eye irritant ingested	Low water solubility Highly volatile Toxic to mammals Toxic to aquatic organisms Possible carcinogen Slightly toxic if ingested	Acute Tox. 4 (100%)		H302 (100%): Harmful if swallowed [Warning Acute toxicity, oral]	Acute Tox. 4 Acute Tox. 4 Aquatic Chronic 2
50471-44-8	XXX	Vinclozolin							1	Low solubility in water Non-persistent in soils & water mobile (Lacewings)-<i>Chrysoperla carnea</i> Toxic to aquatic organisms Endocrine issues - Competitive binding to androgen receptor Reproductive / developmental toxicant Skin sensitizer carcinogen gland toxicant- androgen US EPA - possible human carcinogen; CLP data - suspected carcinogen	Low volatility Moderately harmful to Beneficial insects EDC - Skin irritant Possible Renal and prostate	Skin Sens. 1 (100%) Carc. 2 (100%) Repr. 1B (100%) Aquatic Chronic 2 (100%)		H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H351 (100%): Suspected of causing cancer [Warning Carcinogenicity] H360 (100%): May damage fertility or the unborn child [Danger Reproductive toxicity] H411 (100%): Toxic to aquatic life with long lasting effects [Hazardous to the aquatic environment, long-term hazard]	Skin Sens. 1 H317 Carc. 2 H351 Aquatic Chronic 2 H411 Repr. 1B H360FD
12122-67-7		Zineb							1	Chemical subject to PIC regulations - Low solubility in water Low volatility Non-persistent in soil & water Slightly mobile in soil Low leachability Medium potential for particle bound transport Toxic to earthworms & honey bees Harmful to beneficial insects (predatory mites) - <i>Typhlodromus pyri</i> Toxic to aquatic organisms		Skin Sens. 1 (100%) STOT SE 3 (100%)		H317 (100%): May cause an allergic skin reaction [Warning Sensitization, Skin] H335 (100%): May cause respiratory irritation [Warning Specific organ toxicity, single exposure; respiratory tract irritation]	Skin Sens. 1 H317 STOT SE 3 H 335

UnPoison South Africa's Highly Hazardous Pesticide Database vs EU's According to JMPM Criteria (FAO/WHO), GHS and ECHA Classification Systems - February 2025

	Zinc phosphide								Low solubility in water Non-persistent in soil & birds Toxic to fish Highly toxic to algae Unspecified genotoxicity type Possible Acetyl cholinesterase inhibitor phosphine form May be fatal if swallowed	Low volatility Highly toxic to mammals Genotoxic- Reproductive / developmental Possible Acetyl cholinesterase inhibitor Highly toxic in phosphine form	Water-react. 1 (100%) Acute Tox. 2 (100%) Acute Tox. 3 (36.4%) Acute Tox. 4 (49.4%) Aquatic Acute 1 (100%) Aquatic Chronic 1 (/100%).		H260 (100%): In contact with water releases flammable gases which may ignite spontaneously [Danger Substances and mixtures which in contact with water, emit flammable gases] H300 (100%): Fatal if swallowed [Danger Acute toxicity, oral]	Water-react. 1 H260 Acute Tox. 2 H300 Aquatic Acute 1 H400 Aquatic Chronic 1 H410
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TOTALS	35	24	12	45	15	40	3	194
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Note: This list does not identify partially banned, restricted use, substances of concern, or pesticides to be banned in 2025. Only pesticides that are completely banned with no permitted legal application at the time of publishing. Additionally, there may be other registered HHP's and coformulants that UnPoison has not included due to lack of access to data as the national pesticide registry is not public.

* - Data obtained from ECHA and C&L INVENTORY & ECHACHEM

- Data obtained from PubChem and PPDB

* - Chromium data obtained from FEDRIP, 2008

Sources/References:

- NIH. National Library of Medicine. National Centre for Biotechnology Information. PUBCHEM online data source. Available at <https://pubchem.ncbi.nlm.nih.gov/>
- European Chemicals Agency. C&L Inventory of chemicals. Online data tool. Available at: https://echa.europa.eu/information-on-chemicals/cl-inventory-database?p_p_id=disccinventory_WAR_disccinventoryportlet&p_p_lifecycle=0&p_p_state=normal&p_p_mode=view&p_p__state=normal
- University of Hertfordshire. Pesticide Properties Database. An online chemicals database housing ecotoxicity and environmental fate data as well as human health data. Available at: <https://item.herts.ac.uk/aeru/ppdb/>
- EFSA. European Food Safety Authority was contacted for risk assessments on the pesticides in the database and is available at: <https://www.efsa.europa.eu/en>
- The CAS registry hosted by CAS.org was used to verify the CAS numbers and is available at: <https://www.cas.org/cas-data/cas-registry>